TRANSCRIPTOMICS AND RNA-SEQ

Molecular Principles, Experimental Design and Quantitative Analysis

GENOME | TRANSCRIPTOME | RNA-SEQ | DATA ANALYSIS | BIOLOGICAL INSIGHTS

SAMPLE COLLECTION → RNA EXTRACTION → LIBRARY PREPARATION → SEQUENCING → BIOINFORMATICS ANALYSIS → DIFFERENTIAL EXPRESSION

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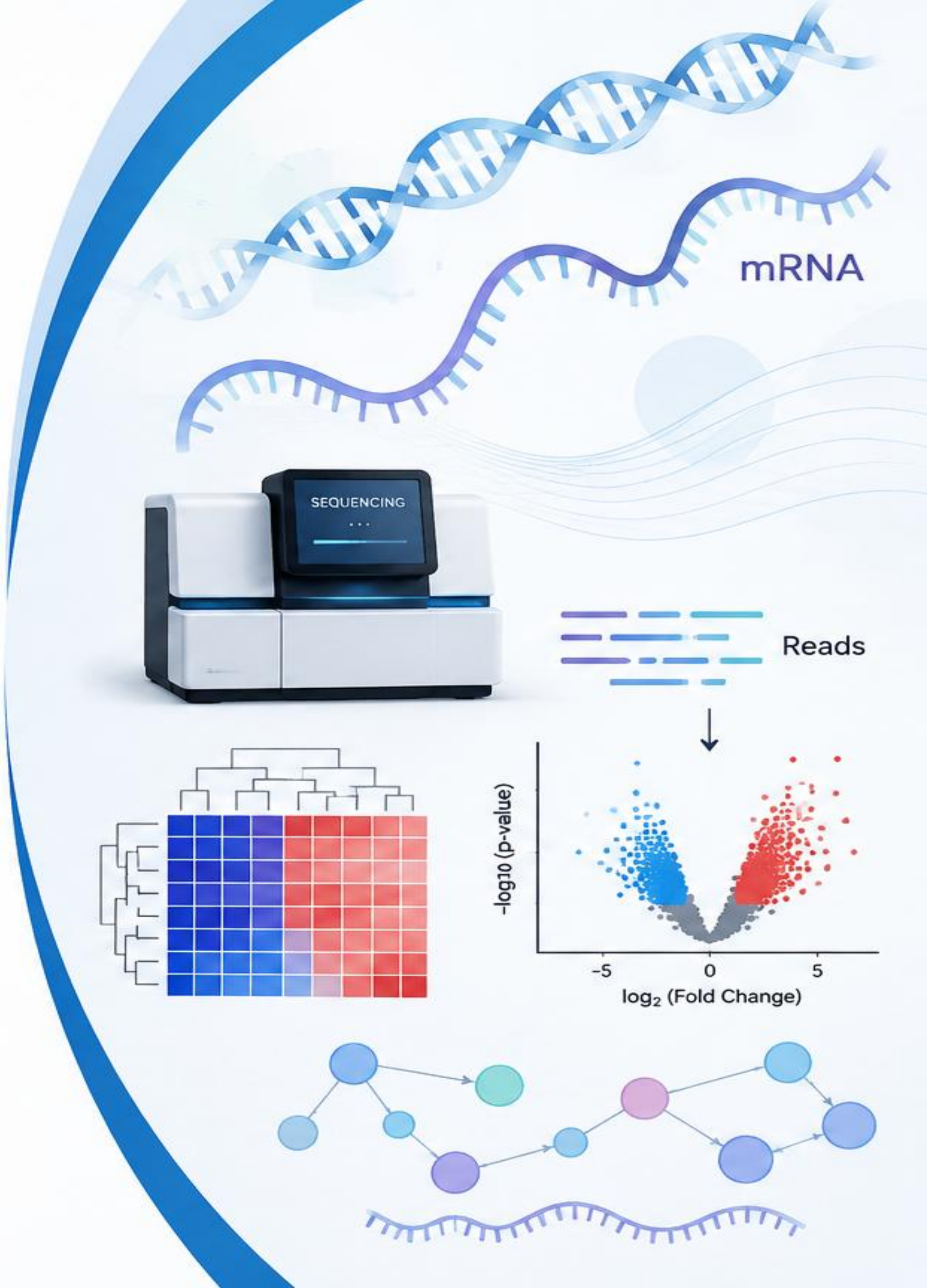
LEARNING OBJECTIVES

By the end of this session, you will be able to:

- Understand the basic concepts and significance of **transcriptomics**.
- Explain **gene expression** and the central dogma at the molecular level.
- Differentiate between major types of **RNA-seq** technologies and their applications.
- Describe key steps in an RNA-seq **experimental design** and data generation workflow.
- Outline the basics of transcriptomics **data analysis** and interpretation.
- Appreciate the **challenges, limitations**, and **future perspectives** in transcriptomics.



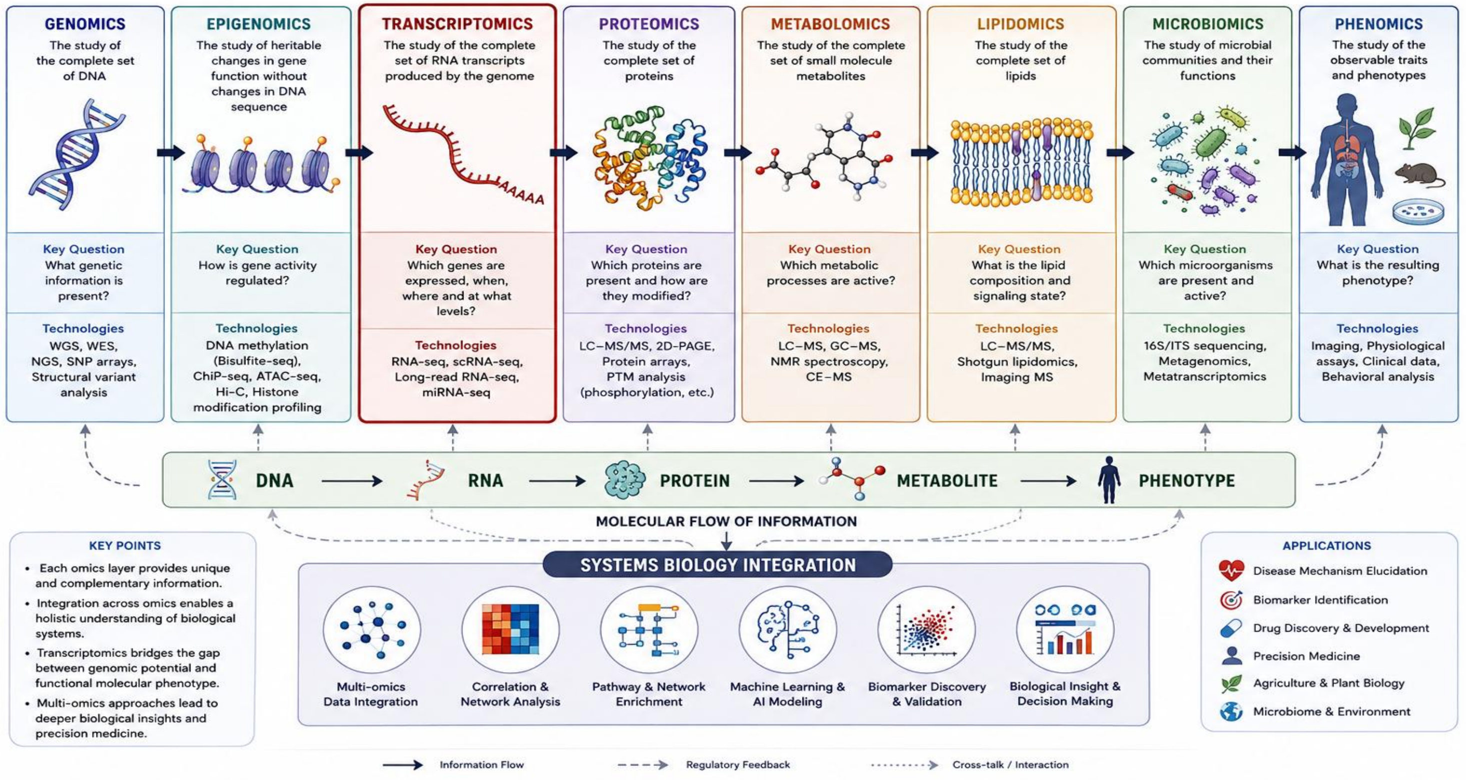
THE OMICS LANDSCAPE

From Genome to Phenome: An Integrated View of Biology

- **Omics** refers to the comprehensive, high-throughput study of biological molecules at a particular molecular level.

The major omics disciplines include:

- **Genomics** → DNA
- **Epigenomics** → epigenetic modifications
- **Transcriptomics** → RNA
- **Proteomics** → Proteins
- **Metabolomics** → Small-molecule metabolite
- **Lipidomics** → Lipids
- **Glycomics** → Carbohydrates/glycans
- **Microbiomics** → Microbial communities
- **Phenomics** → Observable traits



Central concept : Genotype → Molecular State → Phenotype

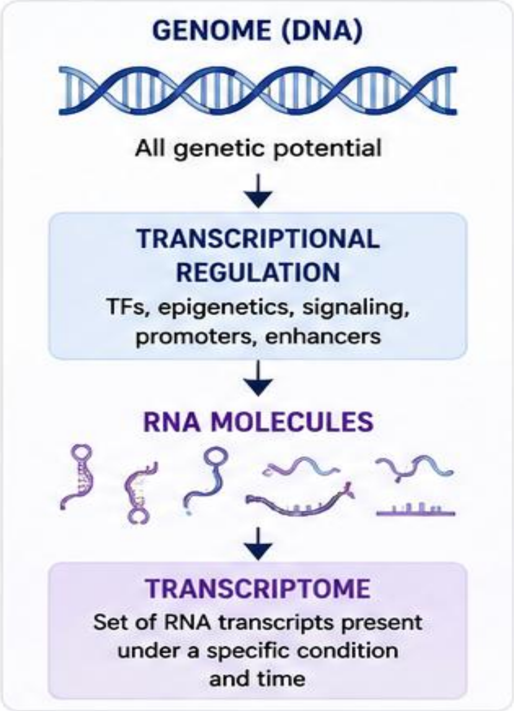
What is the Transcriptome?

Genome = complete DNA information

Transcriptome = RNA molecules expressed by a cell/tissue at a particular time

The transcriptome changes with:

- Development
- Environment
- Disease
- Stress
- Tissue / cell type



GENOME vs TRANSCRIPTOME

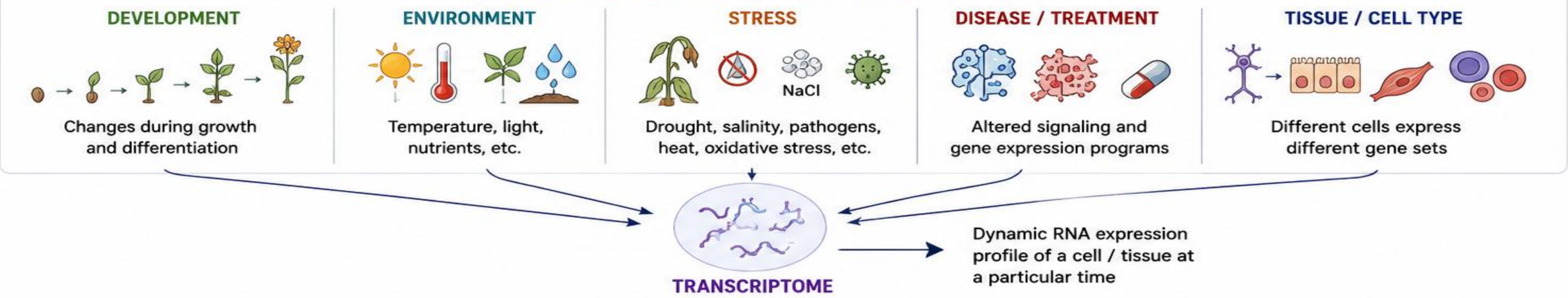
Feature	Genome	Transcriptome
Molecule	DNA	RNA
Nature	Relatively stable	Highly dynamic
Information	Genetic potential	Transcriptional state
Presence	Mostly identical among cells	Varies between cells/tissues
Content	Includes expressed and non-expressed genes	Reflects transcripts actually present
Change over time	Very limited	Rapid and condition-dependent

Key Concept

Same genome ≠ same transcriptome

RNA-seq allows us to capture the transcriptome, a snapshot of gene expression under a given condition.

FACTORS THAT INFLUENCE THE TRANSCRIPTOME



What Constitutes the Transcriptome?

The transcriptome is not simply "**mRNA**."

The transcriptome can include:

- Protein-coding mRNA
- Long non-coding RNA (lncRNA)
- microRNA
- siRNA
- piRNA
- Ribosomal RNA
- Transfer RNA
- Circular RNA
- Antisense transcripts
- Alternative transcript isoforms

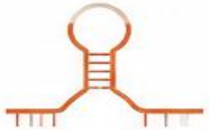
Protein-coding mRNA

Carries information to synthesize proteins.

Long non-coding RNA (lncRNA)

Transcripts >200 nt with regulatory or structural roles, non-protein-coding.

microRNA (miRNA)



~22 nt small RNAs that regulate gene expression post-transcriptionally.

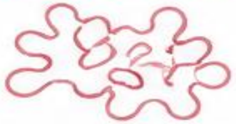
siRNA

~21–23 nt double-stranded RNAs that mediate RNA interference and gene silencing.

piRNA

~24–31 nt RNAs that interact with PIWI proteins and silence transposons, mainly in germ cells.

Ribosomal RNA (rRNA)



Structural and catalytic core components of ribosomes.

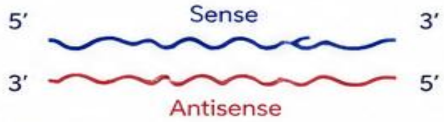
Transfer RNA (tRNA)

Adaptor RNAs that bring amino acids to the ribosome during translation.

Circular RNA (circRNA)

Covalently closed loop RNAs with regulatory functions.

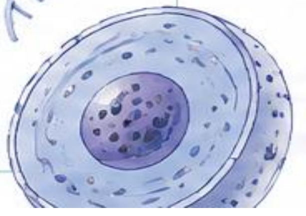
Antisense transcripts

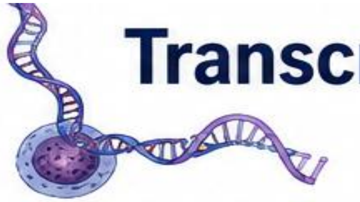


Transcripts from the opposite strand that can regulate gene expression.

Alternative transcript isoforms

Different combinations of exons (via alternative splicing, alternative promoters, or polyadenylation) produce multiple transcript isoforms from the same gene.





Transcriptome as a Dynamic Molecular Phenotype

The transcriptome is not static — it changes continuously as genes are turned on, transcripts are produced, and RNA molecules are degraded.

Key Concept

The level of any transcript at a given moment is the result of a balance between its synthesis (transcription) and its removal (degradation).

THE DYNAMIC NATURE OF THE TRANSCRIPTOME

Unlike the genome, which is relatively constant (except for mutations), the transcriptome varies with:

- Developmental stage
- Cell type and identity
- Environmental cues (e.g., stress, nutrients, light)
- Circadian rhythm and time of day
- Disease and physiological state
- Signaling pathways and stimuli

Thus, the transcriptome is a molecular phenotype that reflects the current state of a cell or organism.

Mathematical Representation

For a gene g, the rate of change in the abundance of its mature transcript M_g can be described as:

$$\frac{dM_g}{dt} = k_{transcription} - k_{degradation}$$

where:

M_g = abundance (molecules or concentration) of transcript from gene g

$k_{transcription}$ = transcription rate (synthesis of new RNA molecules)

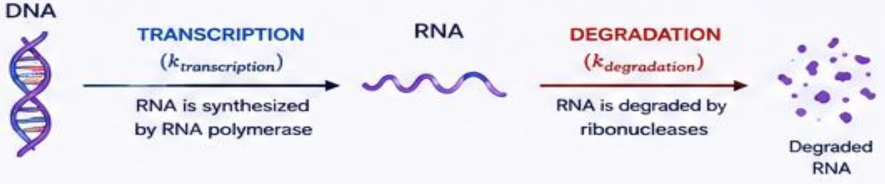
$k_{degradation}$ = degradation rate (loss of RNA molecules)

At steady state $(dM_g/dt = 0)$:

$$M_g = \frac{k_{transcription}}{k_{degradation}}$$

The observed RNA level reflects the ratio of production to degradation.

THE LIFE CYCLE OF AN RNA MOLECULE



Factors influencing $k_{transcription}$ and $k_{degradation}$

Increase transcription:

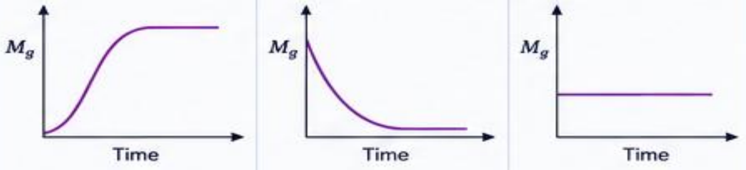
- Active promoters and enhancers
- Transcription factors
- Open chromatin
- Signaling activation

Increase degradation:

- RNA-binding proteins
- miRNAs and siRNAs
- RNA structure and modifications
- Exonucleases (e.g., XRN1, 3'→5')

TRANSCRIPT ABUNDANCE OVER TIME

Different kinetic scenarios for a gene g



RNA level increases until a new steady state.

RNA level decreases to a lower steady state.

RNA level remains constant over time.

Cells constantly adjust transcription and degradation rates to fine-tune gene expression.

WHAT DOES MEASURED RNA ABUNDANCE REPRESENT?

Measured abundance (M_g^{meas}) at time t is the result of:

$$M_g^{meas}(t) = f(k_{transcription}(t),\ k_{degradation}(t),\ history)$$

- It is a snapshot of a dynamic process.
- Two genes with the same RNA level may have very different transcription and degradation rates. (e.g., high turnover vs. low turnover RNAs)
- Short half-life RNAs respond quickly to changes.
- Long half-life RNAs integrate signals over time.

Therefore, the transcriptome reflects both the current regulatory inputs and the stability of transcripts.

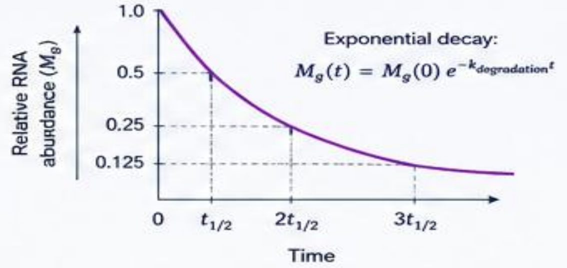
RNA Half-life and Turnover

Half-life $(t_{1/2})$ is the time required for RNA abundance to reduce by 50%.

$$t_{1/2} = \frac{\ln 2}{k_{degradation}}$$

Examples:

- Histone mRNA: ~5–10 min
- c-Myc mRNA: ~20–30 min
- rRNA, tRNA: hours to days

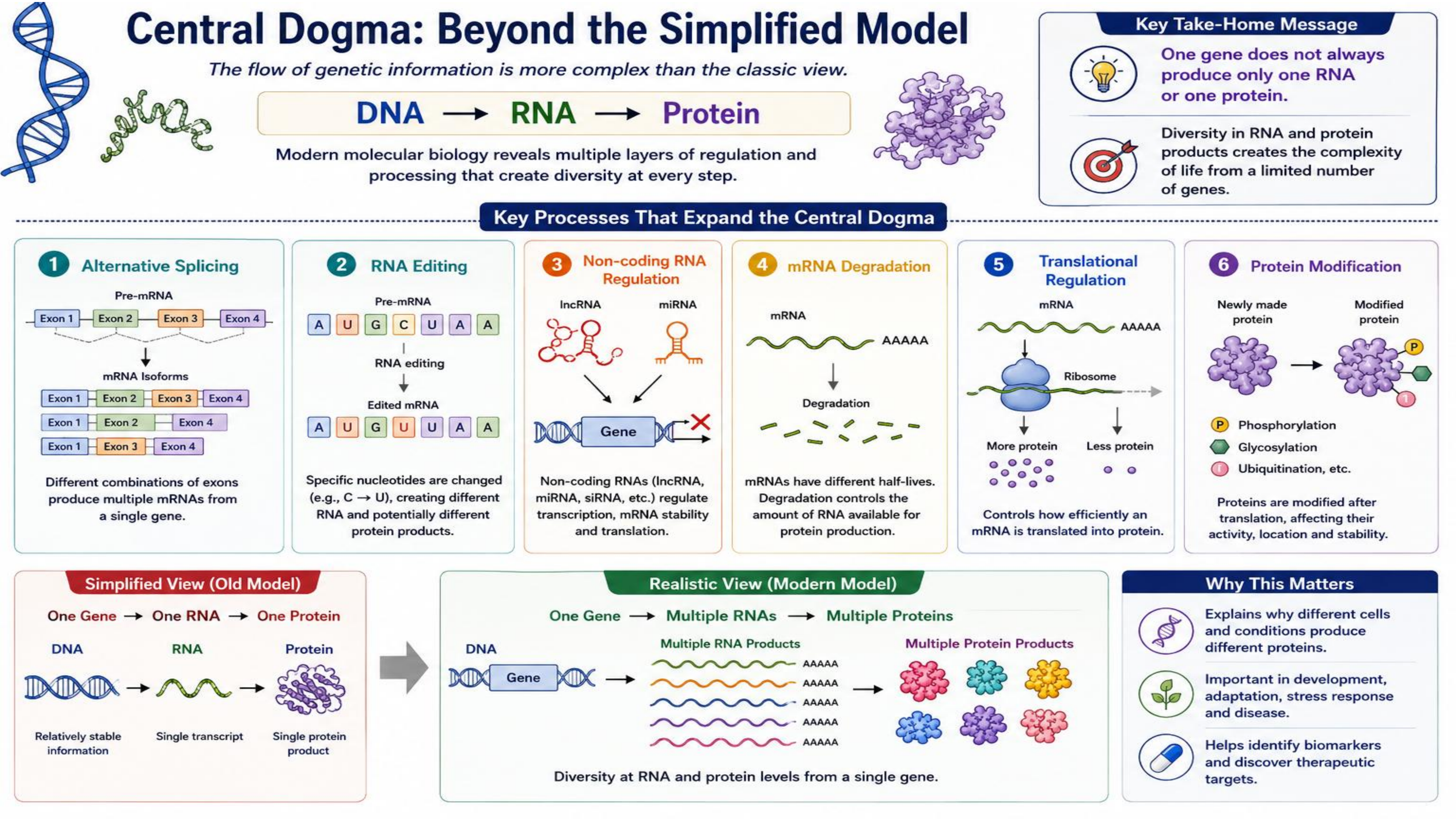


Biological Implication

Dynamic regulation of transcription and RNA stability allows cells to rapidly and precisely control gene expression in response to internal and external signals.

WHY THIS MATTERS

- Interpreting RNA-seq data requires considering RNA stability, not just transcription.
- Therapeutic strategies can target transcription machinery or RNA stability pathways.
- Understanding dynamics helps in deciphering mechanisms of development, stress responses, and disease.
- The transcriptome is a real-time readout of how cells integrate multiple regulatory layers.



Central Dogma: Beyond the Simplified Model
The flow of genetic information is more complex than the classic view.
DNA → RNA → Protein
Modern molecular biology reveals multiple layers of regulation and processing that create diversity at every step.
Key Take-Home Message
One gene does not always produce only one RNA or one protein.
Diversity in RNA and protein products creates the complexity of life from a limited number of genes.
Key Processes That Expand the Central Dogma
1 Alternative Splicing
Pre-mRNA
Exon 1
Exon 2
Exon 3
Exon 4
mRNA Isoforms
Exon 1 Exon 2 Exon 3 Exon 4
Exon 1 Exon 2 Exon 4
Exon 1 Exon 3 Exon 4
Different combinations of exons produce multiple mRNAs from a single gene.
2 RNA Editing
Pre-mRNA
A U G C U A A
RNA editing
Edited mRNA
A U G U U A A
Specific nucleotides are changed (e.g., C → U), creating different RNA and potentially different protein products.
3 Non-coding RNA Regulation
lncRNA
miRNA
Gene
Non-coding RNAs (lncRNA, miRNA, siRNA, etc.) regulate transcription, mRNA stability and translation.
4 mRNA Degradation
mRNA
AAAAA
Degradation
mRNAs have different half-lives. Degradation controls the amount of RNA available for protein production.
5 Translational Regulation
mRNA
AAAAA
Ribosome
More protein
Less protein
Controls how efficiently an mRNA is translated into protein.
6 Protein Modification
Newly made protein
Modified protein
P
P Phosphorylation
Glycosylation
Ubiquitination, etc.
Proteins are modified after translation, affecting their activity, location and stability.
Simplified View (Old Model)
One Gene → One RNA → One Protein
DNA
RNA
Protein
Relatively stable information
Single transcript
Single protein product
Realistic View (Modern Model)
One Gene → Multiple RNAs → Multiple Proteins
DNA
Gene
Multiple RNA Products
AAAAA
AAAAA
AAAAA
AAAAA
AAAAA
Multiple Protein Products
Diversity at RNA and protein levels from a single gene.
Why This Matters
Explains why different cells and conditions produce different proteins.
Important in development, adaptation, stress response and disease.
Helps identify biomarkers and discover therapeutic targets.

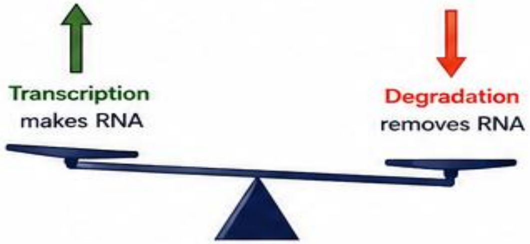
RNA Stability and Degradation

*RNA abundance in a cell is determined not only by **how much** RNA is produced, but also by **how long** it survives.*

What Controls RNA Abundance?

Steady-state RNA level is a balance between

Transcription (production) and
Degradation (removal)



Major Mechanisms of RNA Degradation and Their Regulators

1. RNA-Binding Proteins (RBPs)

RBPs bind to specific sequences or structures in RNA and affect its stability.

Functions:
- Protect RNA from decay
- Recruit decay machinery
- Influence localization
- Control translation

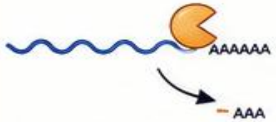
2. miRNAs

miRNAs bind to target mRNAs (usually at 3′ UTR) and either repress translation or promote degradation.

Outcomes:
- Translational repression
- mRNA destabilization
- mRNA degradation

3. Deadenylation

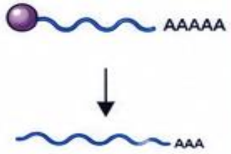
The poly(A) tail is shortened by deadenylase enzymes. This is the first and key step in most mRNA decay pathways.



Effect:
Destabilizes mRNA and promotes further decay.

4. Decapping

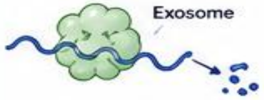
The 5′ cap structure (m^7G) is removed by decapping enzymes.



Effect:
Allows exonucleases to degrade RNA from 5′ end.

5. Exosome-Mediated Degradation (3′ → 5′)

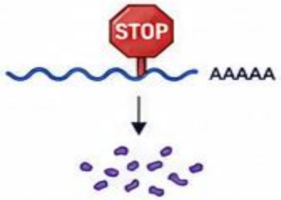
The exosome complex degrades RNA from the 3′ end after deadenylation (and often decapping).



Effect:
Progressive breakdown of RNA to nucleotides.

6. Nonsense-Mediated Decay (NMD)

mRNAs containing premature stop codons are recognized and rapidly degraded.



Effect:
Prevents production of truncated, potentially harmful proteins.

Why RNA Levels Do Not Always Match Transcription Levels

Case 1:
High Transcription
+ Stable RNA

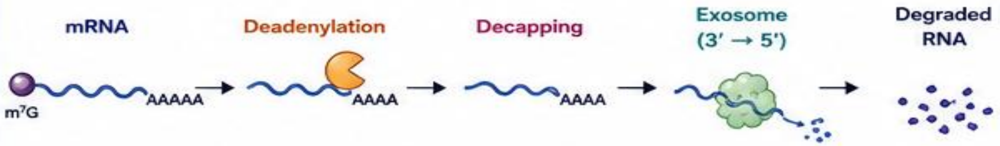
High transcription → Slow degradation → High RNA level

Case 2:
High Transcription
+ Unstable RNA

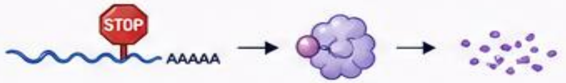
High transcription → Fast degradation → Low RNA level

Therefore, high transcription ≠ necessarily high steady-state RNA abundance.

A Typical mRNA Decay Pathway



Other Pathway: Nonsense-Mediated Decay (NMD)



Triggered by premature stop codons or other abnormal features.

Key Take-Home Message

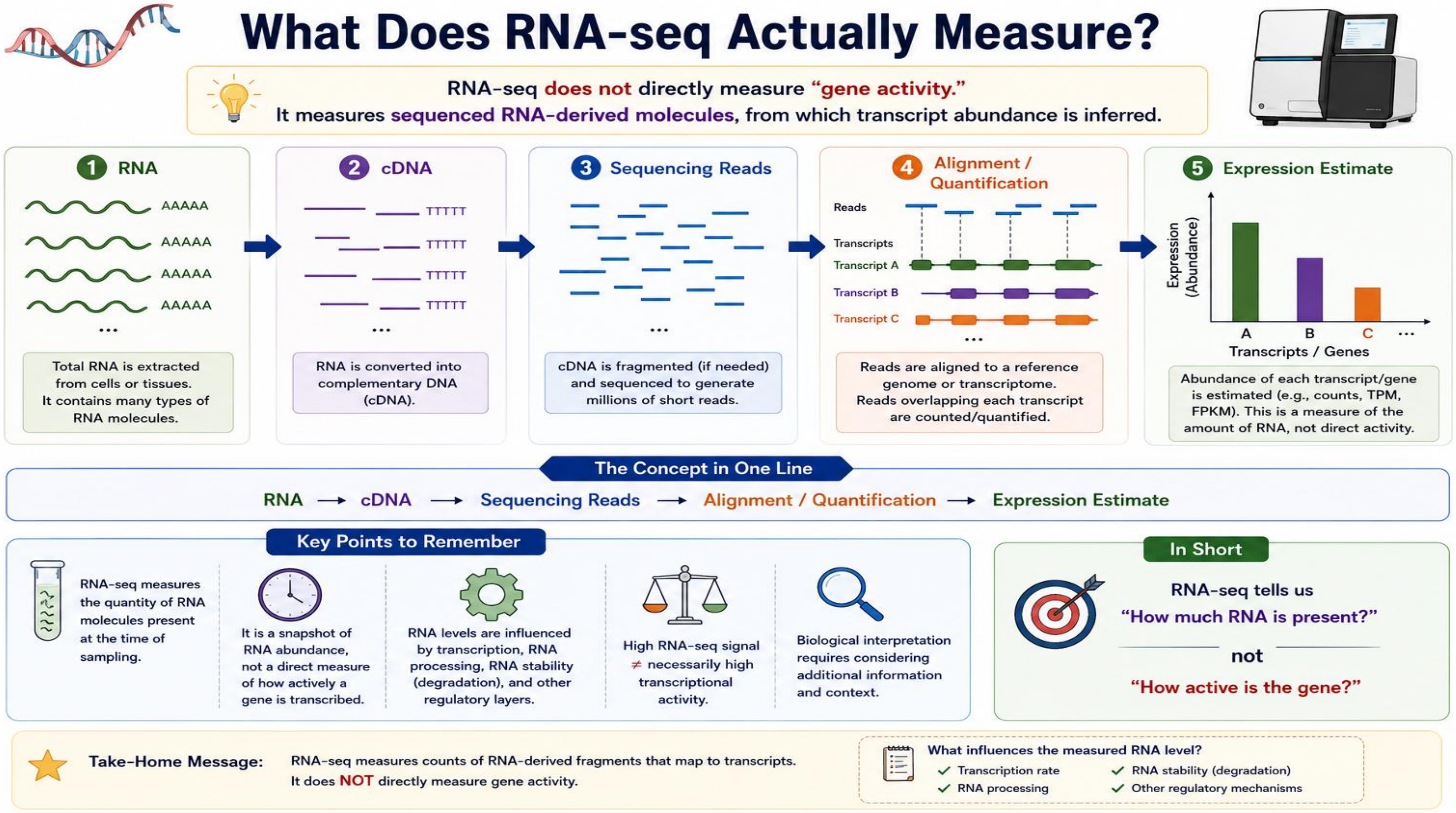
RNA abundance is determined by a dynamic balance between production and multiple degradation pathways.

High transcription
≠
high steady-state RNA abundance.

Biological Significance

- Allows rapid changes in gene expression.
- Ensures removal of faulty or unwanted RNAs.
- Fine-tunes protein production.
- Important in development, stress response, and disease.
- Targeted by many therapeutic strategies and drugs.

Quantitative Transcriptomics



Transcriptome
Sequencing
Expression
Discovery

Different Sequencing Platforms for RNA-seq

Multiple technologies with different read lengths, throughput, accuracy and cost

Same goal
Different approaches
Deeper insights

	Illumina (e.g., NovaSeq, NextSeq, MiSeq)	Thermo Fisher Ion Torrent (e.g., Ion S5, Genexus)	PacBio (e.g., Sequel II, Revio)	Oxford Nanopore Technologies (e.g., MinION, GridION, PromethION)	BGI (e.g., DNBSEQ-T7, MGISEQ)	Sanger Sequencing (e.g., ABI 3730)
Instrument (example)	 Most widely used for RNA-seq	 Semiconductor sequencing (detects pH change)	 Single-molecule real-time (SMRT) sequencing	 Single-molecule sequencing (pore-based, measures electrical signal)	 DNA nanoball (DNB) sequencing	 Chain-termination sequencing (capillary electrophoresis)
Sequencing chemistry	Sequencing by synthesis (reversible terminators)	Semiconductor sequencing (pH detection)	Real-time single-molecule sequencing	Nanopore sequencing (electrical signal)	Sequencing by synthesis (DNB technology)	Sanger sequencing (dideoxy terminators)
Read length	50–300 bp (single-end or paired-end)	100–400 bp (single-end)	10–25 kb (HiFi reads) (up to >100 kb)	Up to >1 Mb (typical 1–100 kb)	50–400 bp (single-end or paired-end)	~700–1,000 bp
Throughput (per run)	Millions to billions of reads (up to ~6 Tb per run)	Up to ~80 million reads (per run, platform dependent)	~5–100 million reads (platform dependent)	Millions of reads (up to >100 Gb per run)	Up to ~6 Tb per run (platform dependent)	Hundreds of reads per run
Accuracy	High (≥99.9%)	~98–99%	HiFi reads ≥99.5% (long reads)	~95–99% (improving with newer chemistries)	High (≥99.9%)	Very high (>99.99%)
Run time	~4 hours – 3 days (depending on instrument)	~2–4 hours	~10–30 hours	Real-time (minutes to hours)	~12 hours – 3 days	~1–3 hours
Typical applications	Gene expression, differential expression, isoform analysis, single-cell RNA-seq	Targeted RNA-seq, small transcriptome studies, custom panels	Full-length transcripts, isoform discovery, novel transcript detection	Full-length RNA, direct RNA-seq, isoform analysis, novel transcripts, field and real-time applications	Large-scale transcriptome studies, population-scale RNA-seq, clinical research	Validation of RNA-seq results (e.g., qPCR targets, isoforms)
Advantages	High accuracy, high throughput, cost-effective, broad applications	Fast run time, simple workflow, scalable, no optics	Long reads, accurate isoforms (HiFi), no amplification bias	Very long reads, portable devices, real-time analysis, direct RNA-seq (without cDNA)	High throughput, cost-effective (at large scale), high accuracy	Very high accuracy, longer reads than short-read NGS, simple analysis
Limitations	Short reads (may complicate isoform resolution), GC bias	Higher indel error rate (homopolymers), lower throughput than Illumina	Higher cost, lower throughput than short-read platforms	Higher error rate (especially indels), requires higher computational correction	Less widely used globally compared to Illumina, limited long-read options	Low throughput, not suitable for whole-transcriptome analysis
	Current standard for RNA-seq studies	**Useful for targeted and smaller projects**	**Best for full-length transcript isoform analysis**	**Ideal for long-read and real-time RNA-seq**	**Widely used in Asia and growing globally**	**Used for validation, not for large-scale RNA-seq**

Key Takeaways

- Multiple sequencing platforms are available for RNA-seq, each with unique strengths and limitations.
- Illumina is the most widely used due to its high accuracy and throughput.
- Long-read platforms (PacBio, ONT) enable full-length transcript and isoform analysis.
- Platform choice depends on study goals, budget, required read length, and throughput.

Choosing the Right Platform

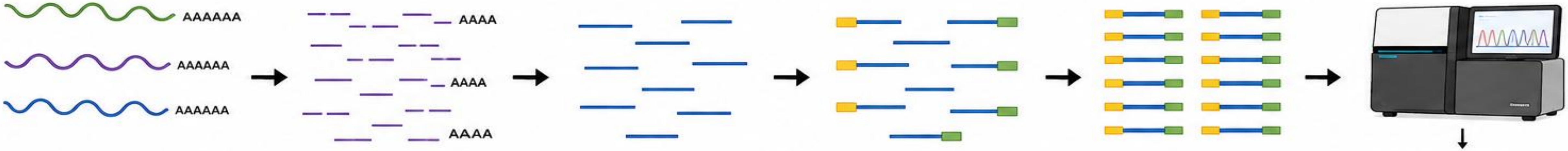
- High-throughput, cost-effective → Illumina / BGI
- Full-length isoforms → PacBio / Oxford Nanopore
- Targeted or small-scale → Ion Torrent
- Validation → Sanger sequencing

Different technologies, a common goal — understanding the transcriptome.

RNA-seq Read Generation: From RNA to Sequencing Reads

Each sequencing read represents a **sampled fragment** from the RNA population.

1 Total RNA | **2 Fragmentation** | **3 cDNA Synthesis** | **4 Adapter Ligation** | **5 Library Amplification** | **6 Sequencing**



ATGCGATACGTTAGCT...
GCCATGTTAGCTACGA...
TTGCAATGCCGATGTA...
...

1 Total RNA

- RNA is extracted from cells or tissues.
- Contains mRNA, rRNA, tRNA, and other RNAs.

2 Fragmentation

RNA is fragmented into short pieces to enable efficient sequencing.

3 cDNA Synthesis

RNA fragments are reverse-transcribed into complementary DNA (cDNA).

4 Adapter Ligation

Adapters (colored sequences) are ligated to both ends of cDNA fragments.

5 Library Amplification

Fragments are amplified by PCR to enrich the library molecules.

6 Sequencing

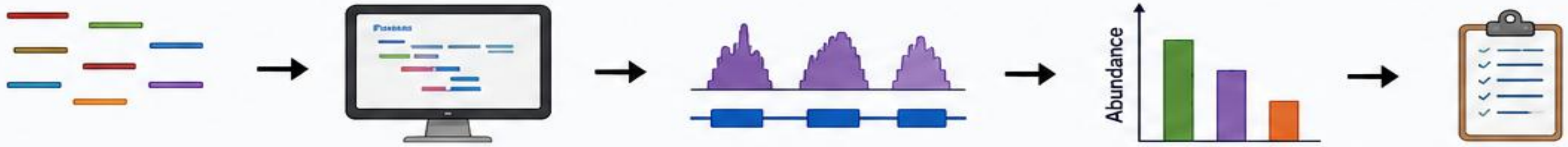
The library is loaded onto a sequencing platform to generate millions of short reads.

Key Idea

RNA-seq does not sequence entire RNAs. It sequences many short fragments.

Each read is a sampled fragment that represents the RNA population in the sample.

What Happens to Each Read?



Sequencing reads are generated → Reads are aligned to a reference genome or transcriptome → Reads overlapping each transcript are counted → Counts are normalized to estimate transcript abundance → Expression levels are used for downstream biological interpretation

Take-Home Message

RNA-seq reads come from fragmented, reverse-transcribed RNA molecules.

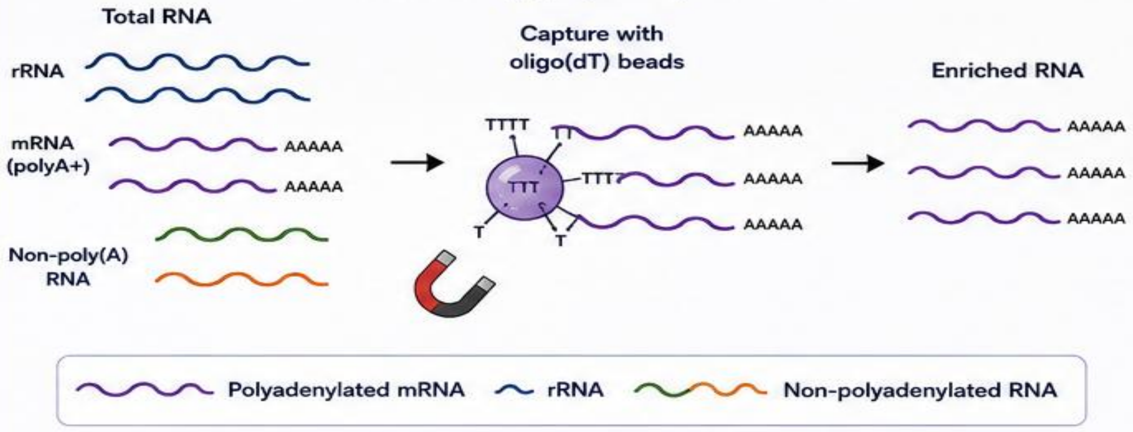
They are not the original RNA, but short pieces that allow us to quantify RNA abundance.

RNA Selection Strategies in RNA-seq

Before sequencing, RNA is selected or depleted to enrich for transcripts of interest.
The two most common approaches are **Poly(A) Selection** and **rRNA Depletion**.

1 Poly(A) Selection

Enriches mainly polyadenylated RNA.



Advantages

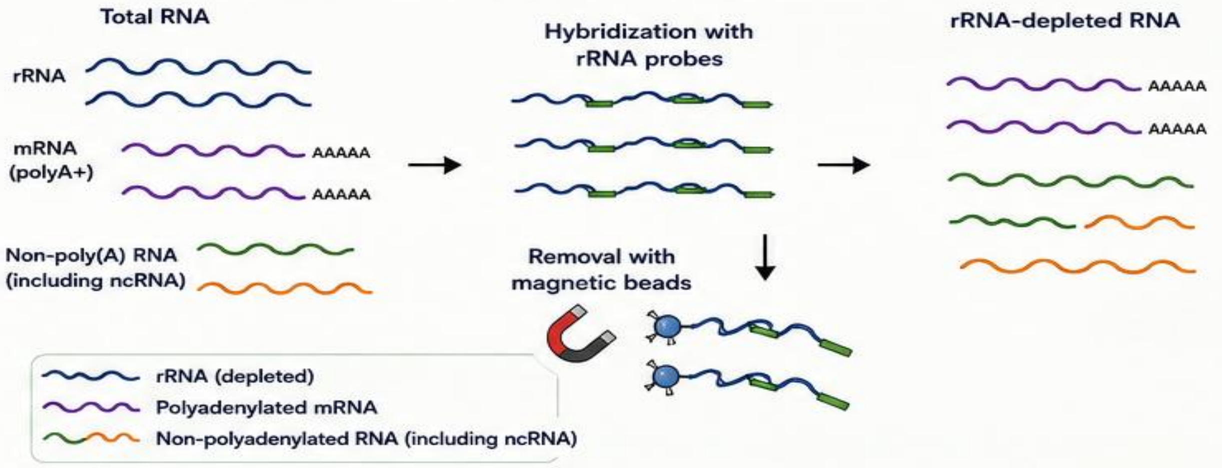
- Strong mRNA enrichment
- Lower rRNA contribution
- Higher sequencing efficiency for coding transcripts

Limitations

- Misses many non-polyadenylated transcripts (e.g., many lncRNAs, histone mRNAs, some snRNAs and snoRNAs, etc.)

2 rRNA Depletion

Removes ribosomal RNA.



Advantages

- Broader transcriptome representation
- Captures poly(A)+ and non-poly(A) RNAs
- Better for non-coding RNA studies
- More complete view of the transcriptome

Limitations

- Greater sequencing complexity
- More residual rRNA may remain compared to poly(A) selection
- Requires more sequencing depth for same coverage

At a Glance

Feature	Poly(A) Selection	rRNA Depletion
Target	Polyadenylated RNA (mainly mRNA)	All RNA except rRNA
Best for	Coding transcript (mRNA) studies	Coding + non-coding RNA studies
rRNA content	Very low	Low to moderate (may remain)
Transcriptome coverage	Narrower	Broader
Sequencing complexity	Lower	Higher

Choice depends on your research question:

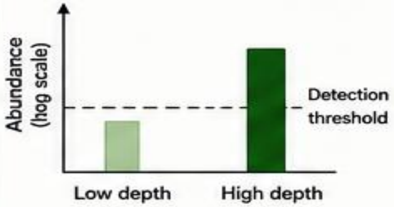
- Focus on mRNA? → Use Poly(A) Selection
- Need the whole picture? → Use rRNA Depletion

Sequencing Depth

Sequencing depth = number of **reads** generated per sample.

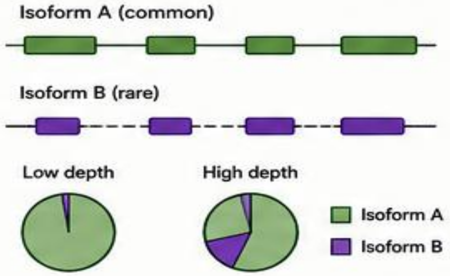
Greater depth improves our ability to detect:

Low-abundance transcripts



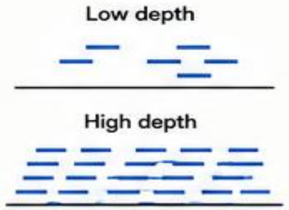
More reads help detect transcripts present at very low levels.

Rare isoforms



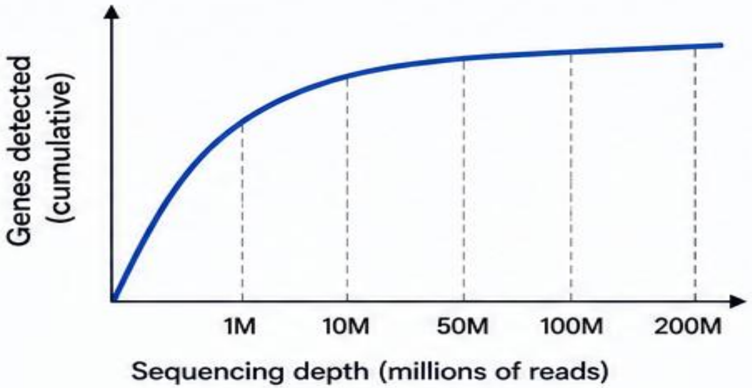
Deeper sequencing increases the chance of capturing rare isoforms.

Weakly expressed genes



More reads provide better quantification of weakly expressed genes.

Increasing depth shows diminishing returns



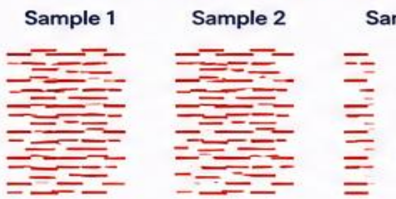
- As depth increases, more genes are detected...
- But after a point, the increase becomes minimal.
- Choose depth based on your biological question, not "more is always better".

But: More reads **do not** compensate for poor biological replication.

Biological replicates capture **biological variability**.
Sequencing depth only reduces **technical noise**.

Example: Depth vs Replication

Many reads, no replication



Problem: Cannot estimate biological variability. Results may be misleading.

Moderate reads, good replication

Better: Captures biological variability. More reliable conclusions.

Take-Home Message

- Sequencing depth improves detection of low-abundance transcripts, rare isoforms, and weakly expressed genes.
- More reads help—but have diminishing returns.
- **Biological replication is irreplaceable.** It is the foundation of trustworthy results.

Key Experimental-Design Principle: Invest in good biological replication first; increase sequencing depth second.

Read Alignment

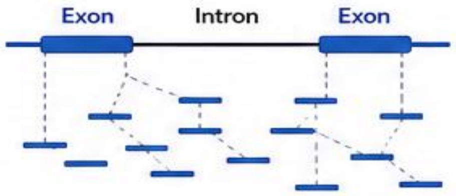
Reads generated by RNA-seq are **aligned** to a reference to determine their origin.

1. Alignment to Reference Genome

Reads are mapped to the entire genome sequence.

Useful for:

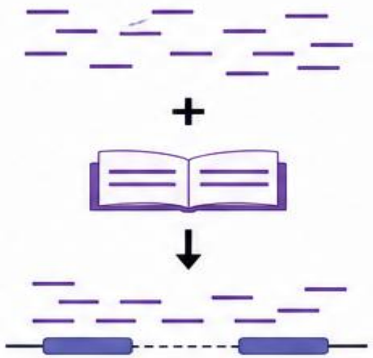
- Novel transcripts
- Splicing events
- Genomic context (e.g., introns, intergenic regions)



Captures reads spanning exon—exon junctions and novel regions.

CONCEPT

Reads + Reference → Alignment

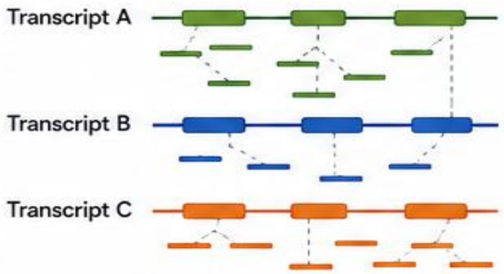


2. Alignment to Reference Transcriptome

Reads are mapped to known transcript sequences.

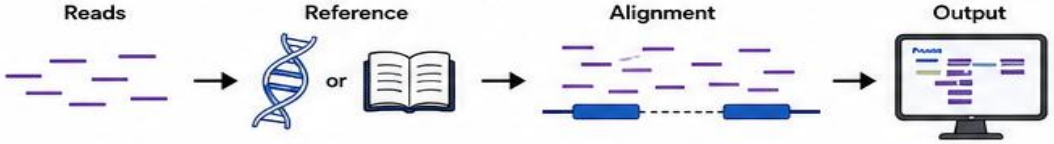
Useful for:

- Transcript quantification
- Known transcript analysis
- Comparisons across samples and conditions



Focuses on known transcripts for accurate quantification.

How Alignment Works (Simplified)



Reads: Millions of short reads from RNA-seq.

Reference: Reference genome or reference transcriptome.

Alignment: Reads are placed at the best matching location.

Output: Aligned reads are used for downstream analyses (e.g., quantification).

Key Takeaway

- Genome alignment = discovery + context (novel transcripts, splicing, genomic location)
- Transcriptome alignment = quantification + efficiency (known transcripts, faster, simpler)
- Both have their place—choice depends on your research question.

Why It Matters

- Accurate alignment is the foundation of reliable expression analysis.
- The choice of reference (genome vs transcriptome) impacts what you can discover and how you interpret your data.
- Good alignment → good biology!

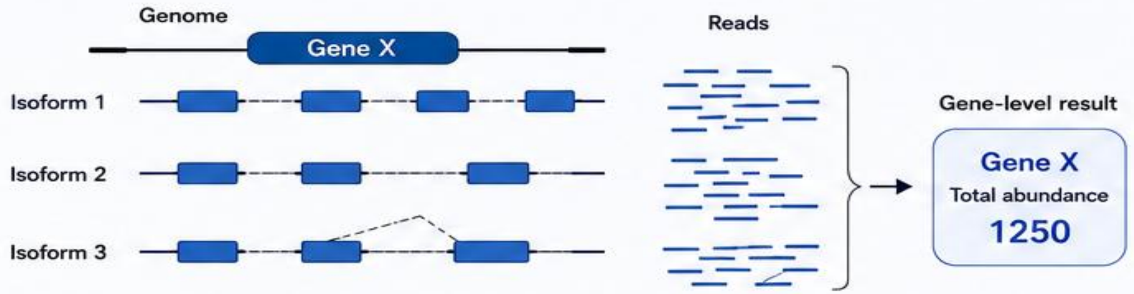
In summary: Reads + Reference → Alignment → Information | Alignment turns raw reads into biological insight.

Gene-Level vs Transcript-Level Quantification

RNA-seq reads can be summarized at **different levels.**
The choice of level determines the **biological insight** you can obtain.

1 Gene-Level Quantification

Gene → Total transcript abundance



What it tells us

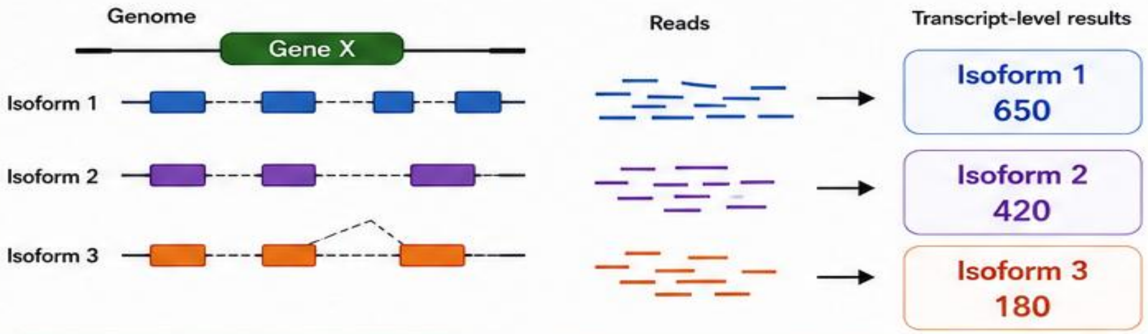
- How much total RNA comes from Gene X.

Limitation

- Cannot distinguish between isoforms.
- Isoform switching or usage changes are hidden.

2 Transcript-Level Quantification

Gene → Isoform_1 + Isoform_2 + Isoform_3



What it tells us

- Abundance of each isoform individually.
- Allows detection of isoform switching and usage changes.

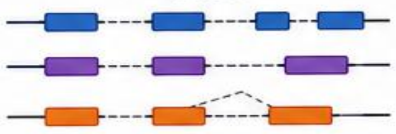
Advantage

- More detailed and biologically informative.
- Reveals alternative splicing and isoform dynamics.

Why It Matters: An Example

Condition A

Gene X



Transcript-level

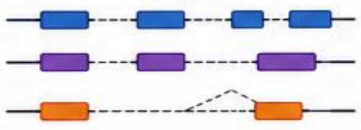
Isoform 1	600
Isoform 2	100
Isoform 3	50

Gene-level

Total = 750

Condition B

Gene X



Transcript-level

Isoform 1	50
Isoform 2	500
Isoform 3	200

Gene-level

Total = 750

Interpretation

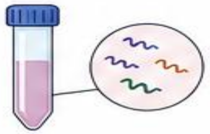
Gene-level analysis suggests no change (750 vs 750).

But transcript-level analysis reveals **a major isoform switch!**

Key Take-Home Message

Gene-level summarizes total signal; transcript-level reveals the biology.
Always consider transcript-level analysis when isoforms matter.

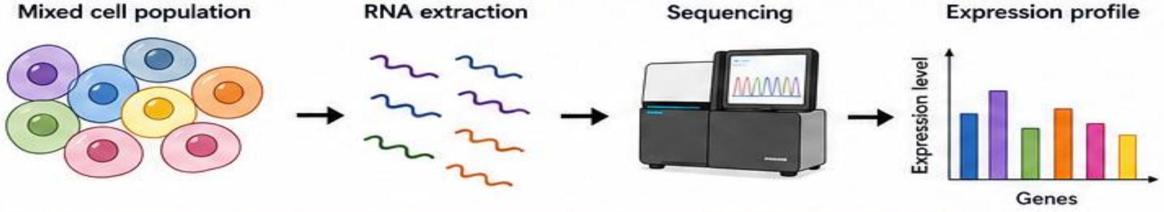
Types of RNA-seq



Bulk RNA-seq

Bulk RNA-seq measures **average** expression across many cells in a sample.

What is Bulk RNA-seq?

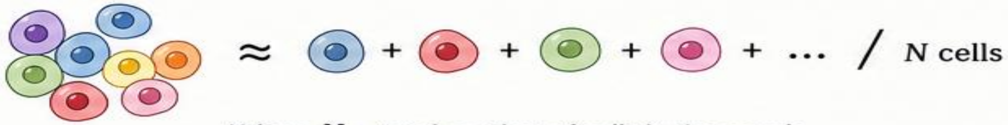


All cells are mixed together.
The measured expression is the **average** across all cells.

Concept

Bulk RNA-seq measures the average expression across many cells.

$$Expression_{bulk} \approx Average\ cellular\ expression$$



Where N = total number of cells in the sample

Strengths

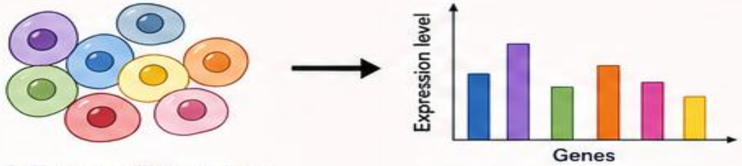
Cost-effective
Many samples can be profiled at a reasonable cost.

High sequencing depth
Allows sensitive detection of low-abundance transcripts.

Mature statistical methods
Well-established tools for analysis, normalization and interpretation.

Key Limitation

Cellular heterogeneity is averaged out.



Different cell types have different expression profiles.

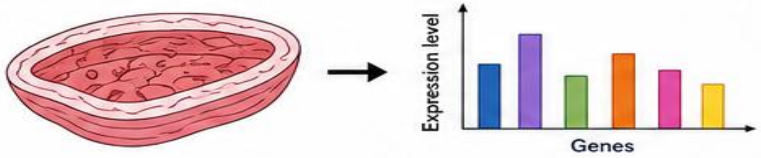
Bulk RNA-seq reports only the average signal.

Important cell-type–specific or rare cell signals **may be hidden.**

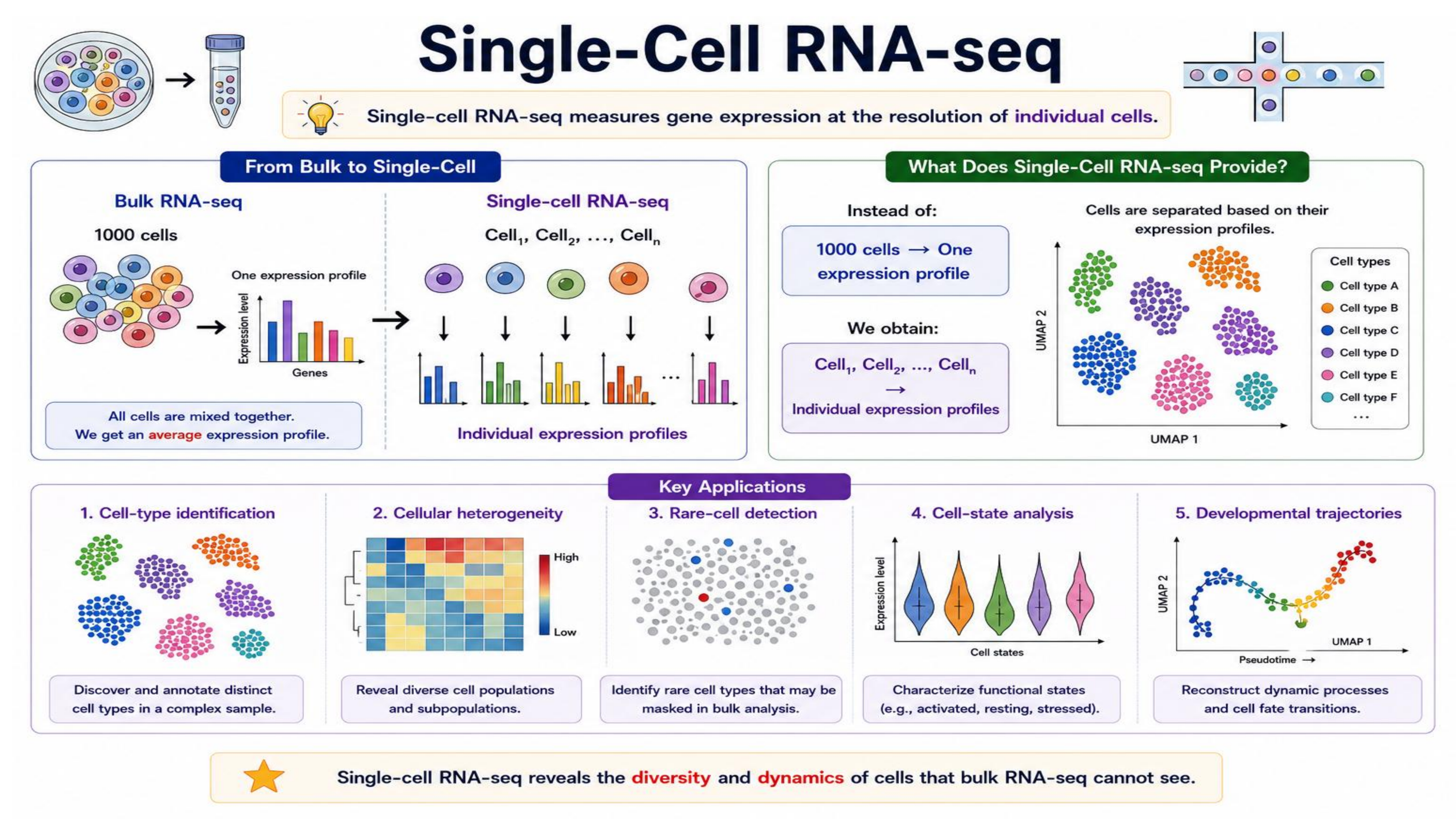
When to Use Bulk RNA-seq?

- When the sample is relatively homogeneous (e.g., purified cells, tissues with few cell types).

- When the goal is to measure overall expression at the tissue or population level.



Bulk RNA-seq is powerful, affordable, and widely used— but it provides an **average view** of a complex mixture of cells.



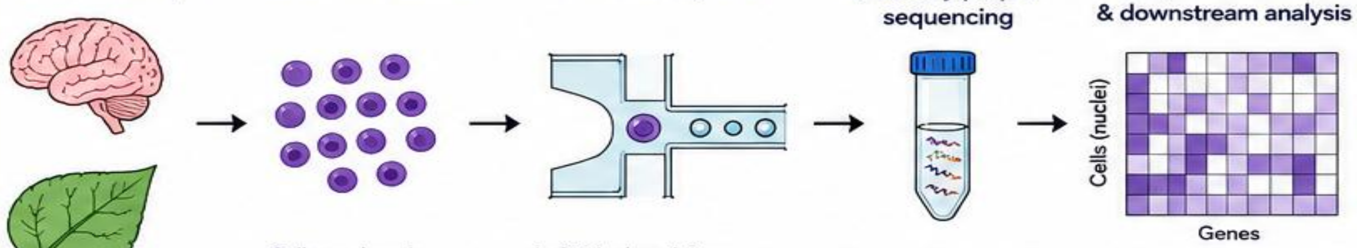
Single-Cell RNA-seq
Single-cell RNA-seq measures gene expression at the resolution of individual cells.
From Bulk to Single-Cell
Bulk RNA-seq
1000 cells
One expression profile
Expression level
Genes
All cells are mixed together.
We get an average expression profile.
Single-cell RNA-seq
$Cell_1$, $Cell_2$, …, $Cell_n$
Individual expression profiles
What Does Single-Cell RNA-seq Provide?
Instead of:
1000 cells → One expression profile
We obtain:
$Cell_1$, $Cell_2$, …, $Cell_n$ → Individual expression profiles
Cells are separated based on their expression profiles.
UMAP 2
UMAP 1
Cell types
Cell type A
Cell type B
Cell type C
Cell type D
Cell type E
Cell type F
…
Key Applications
1. Cell-type identification
Discover and annotate distinct cell types in a complex sample.
2. Cellular heterogeneity
High
Low
Reveal diverse cell populations and subpopulations.
3. Rare-cell detection
Identify rare cell types that may be masked in bulk analysis.
4. Cell-state analysis
Expression level
Cell states
Characterize functional states (e.g., activated, resting, stressed).
5. Developmental trajectories
UMAP 2
UMAP 1
Pseudotime →
Reconstruct dynamic processes and cell fate transitions.
Single-cell RNA-seq reveals the diversity and dynamics of cells that bulk RNA-seq cannot see.

Single-Nucleus RNA-seq

Single-nucleus RNA-seq (snRNA-seq) profiles gene expression from **isolated nuclei** instead of intact cells.

How Single-Nucleus RNA-seq Works

1. Tissue sample
2. Nuclei isolation
3. Nuclei capture
4. Library prep & sequencing
5. Expression matrix & downstream analysis



- Cells are lysed; nuclei are released and purified.
- Individual nuclei are captured (e.g., microfluidics or droplet-based).
- cDNA library preparation and high-throughput sequencing.
- Generate expression matrix and perform downstream analysis.

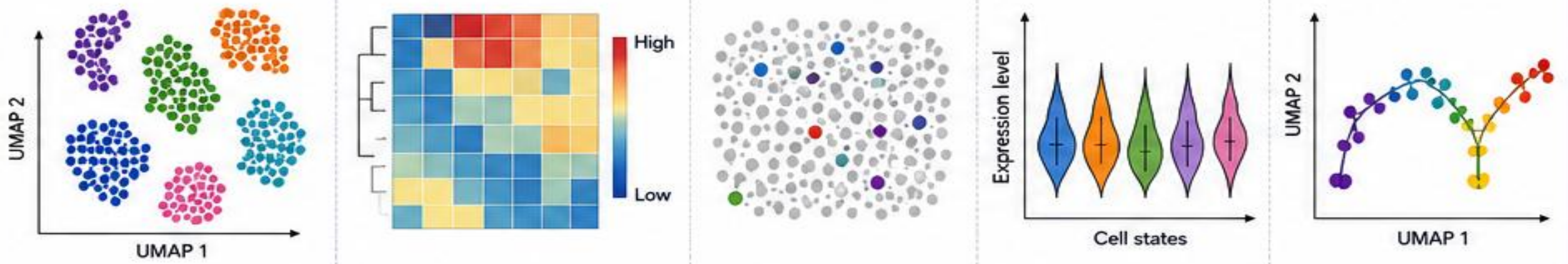
Why Use Single-Nucleus RNA-seq?

Isolated nuclei are more robust than intact cells.

Frozen tissues	Nuclei remain intact after freeze–thaw, while whole cells often lyse.
Difficult-to-dissociate tissues	Dense or connective tissues that are hard to dissociate into viable single cells.
Archived samples	Works with formalin-fixed or stored tissues.
Certain plant tissues	Plant cell walls make cell isolation challenging; nuclei isolation is feasible.

Key Applications

Cell-type identification | Cellular heterogeneity | Rare-cell detection | Cell-state analysis | Developmental trajectories

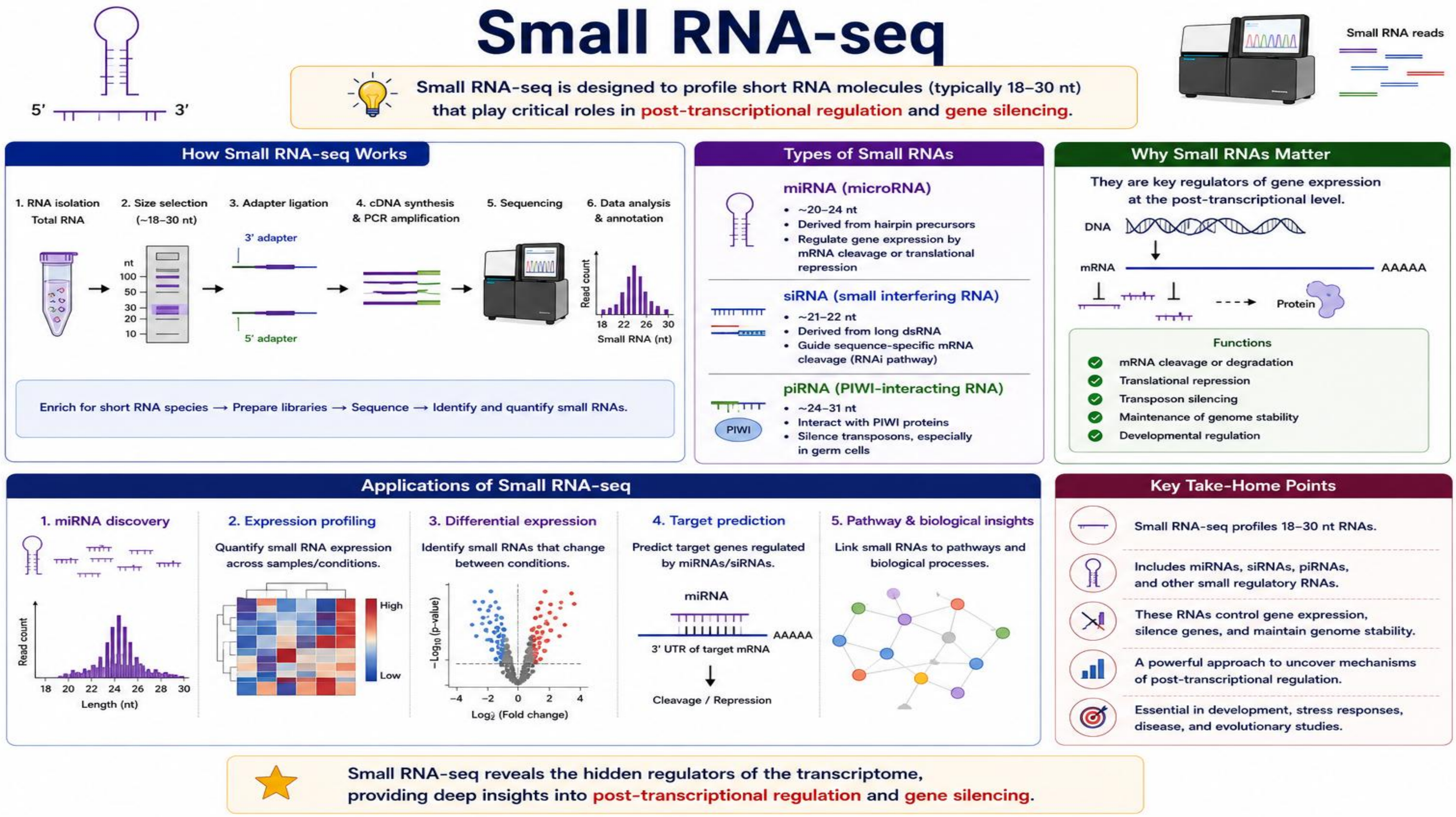


- **Cell-type identification:** Identify distinct cell types from complex tissues.
- **Cellular heterogeneity:** Reveal diverse cell populations and states.
- **Rare-cell detection:** Detect rare or subtle cell populations.
- **Cell-state analysis:** Profile functional states (e.g., activated, resting).
- **Developmental trajectories:** Reconstruct developmental paths and lineage relationships.

snRNA-seq vs scRNA-seq

Feature	scRNA-seq (Whole cells)	snRNA-seq (Nuclei)
Input	Intact viable cells	Isolated nuclei
Best for	Fresh tissues; easy-to-dissociate samples	Frozen, archived, or difficult-to-dissociate tissues
RNA content	Whole transcriptome (cytoplasmic + nuclear)	Primarily nuclear RNA (pre-mRNA + some mRNA)
Sensitivity	Higher for cytoplasmic transcripts	May underestimate some cytoplasmic mRNAs
Practicality	Requires viable, dissociated cells	More robust; works with challenging samples

Key Take-Home Message: Single-nucleus RNA-seq extends single-cell transcriptomics to **challenging samples**, unlocking **insights from tissues and conditions** where single-cell approaches are limited.


Small RNA-seq
5′
3′
Small RNA reads
Small RNA-seq is designed to profile short RNA molecules (typically 18–30 nt)
that play critical roles in post-transcriptional regulation and gene silencing.
How Small RNA-seq Works
1. RNA isolation
Total RNA
2. Size selection
(~18–30 nt)
nt
100
50
30
20
10
3. Adapter ligation
3′ adapter
5′ adapter
4. cDNA synthesis
& PCR amplification
5. Sequencing
6. Data analysis
& annotation
Read count
18 22 26 30
Small RNA (nt)
Enrich for short RNA species → Prepare libraries → Sequence → Identify and quantify small RNAs.
Types of Small RNAs
miRNA (microRNA)
• ~20–24 nt
• Derived from hairpin precursors
• Regulate gene expression by mRNA cleavage or translational repression
siRNA (small interfering RNA)
• ~21–22 nt
• Derived from long dsRNA
• Guide sequence-specific mRNA cleavage (RNAi pathway)
piRNA (PIWI-interacting RNA)
PIWI
• ~24–31 nt
• Interact with PIWI proteins
• Silence transposons, especially in germ cells
Why Small RNAs Matter
They are key regulators of gene expression
at the post-transcriptional level.
DNA
mRNA
AAAAA
Protein
Functions
mRNA cleavage or degradation
Translational repression
Transposon silencing
Maintenance of genome stability
Developmental regulation
Applications of Small RNA-seq
1. miRNA discovery
Read count
18 20 22 24 26 28 30
Length (nt)
2. Expression profiling
Quantify small RNA expression
across samples/conditions.
High
Low
3. Differential expression
Identify small RNAs that change
between conditions.
−Log10 (p-value)
−4 −2 0 2 4
Log2 (Fold change)
4. Target prediction
Predict target genes regulated
by miRNAs/siRNAs.
miRNA
AAAAA
3′ UTR of target mRNA
Cleavage / Repression
5. Pathway & biological insights
Link small RNAs to pathways and
biological processes.
Key Take-Home Points
Small RNA-seq profiles 18–30 nt RNAs.
Includes miRNAs, siRNAs, piRNAs,
and other small regulatory RNAs.
These RNAs control gene expression,
silence genes, and maintain genome stability.
A powerful approach to uncover mechanisms
of post-transcriptional regulation.
Essential in development, stress responses,
disease, and evolutionary studies.
Small RNA-seq reveals the hidden regulators of the transcriptome,
providing deep insights into post-transcriptional regulation and gene silencing.

Long-Read Transcriptomics

Sequencing technologies that read much longer RNA-derived molecules, revealing **full-length transcripts** and **complex transcript structures**.

Major Platforms

PacBio (Iso-Seq, SMRT)

Oxford Nanopore Technologies (cDNA / direct RNA)

Key Advantages

- **Full-length isoforms** — Read from 5′ to 3′ end without assembly.
- **Alternative splicing** — Resolve all splice variants directly.
- **Complex transcript structures** — Detect complex, nested, and overlapping transcripts.
- **Novel isoform discovery** — Identify new transcripts, TSS, and poly(A) sites.

Short Reads vs Long Reads: The Concept

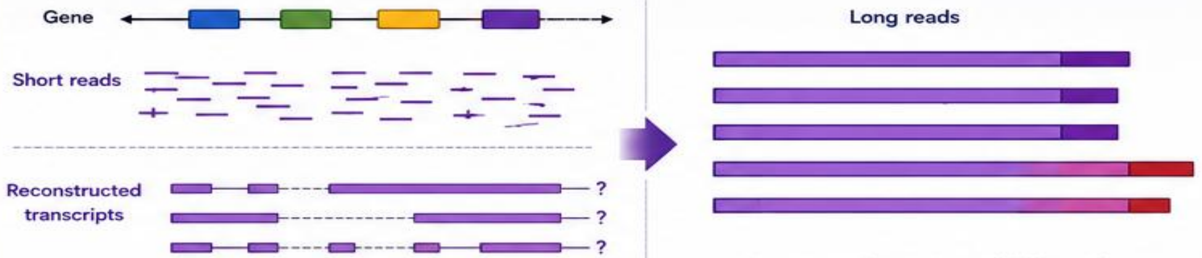
Short reads (e.g., Illumina): Short reads must be assembled computationally to reconstruct transcripts.

- Assembly can be incomplete or chimeric
- Struggles with repetitive or highly similar regions
- Isoform reconstruction can be ambiguous

Long reads (e.g., PacBio, ONT): Long reads observe much of the transcript directly—often the full-length isoform.

- Direct observation of complete transcripts
- Accurate isoform identification
- Better resolution of complex regions

What Long-Read Transcriptomics Enables



Short reads provide pieces...
Long reads provide the picture.

Long reads capture full-length transcripts and resolve isoforms without assembly gaps.

Key Applications

Full-length transcript cataloging
- Build comprehensive reference transcriptomes
- Improve genome annotation

Alternative splicing analysis
- Detect known and novel splice isoforms
- Quantify isoform usage

Complex loci characterization
- Resolve genes with overlapping or nested transcripts
- Handle repetitive regions

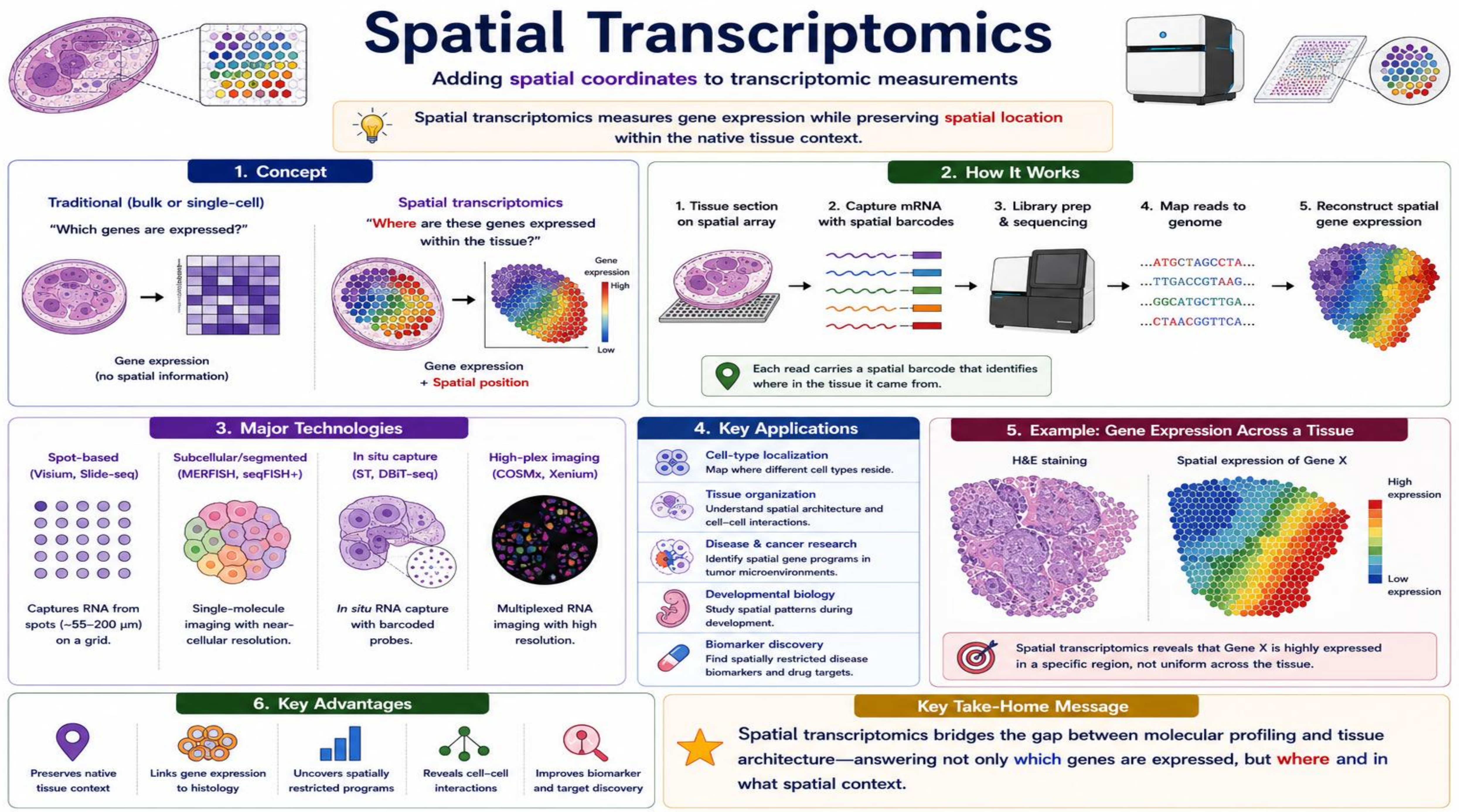
Novel isoform discovery
- Identify new isoforms, novel TSS, and poly(A) sites
- Discover long non-coding RNAs

Better quantification
- More accurate isoform-level quantification
- Improved biological insights

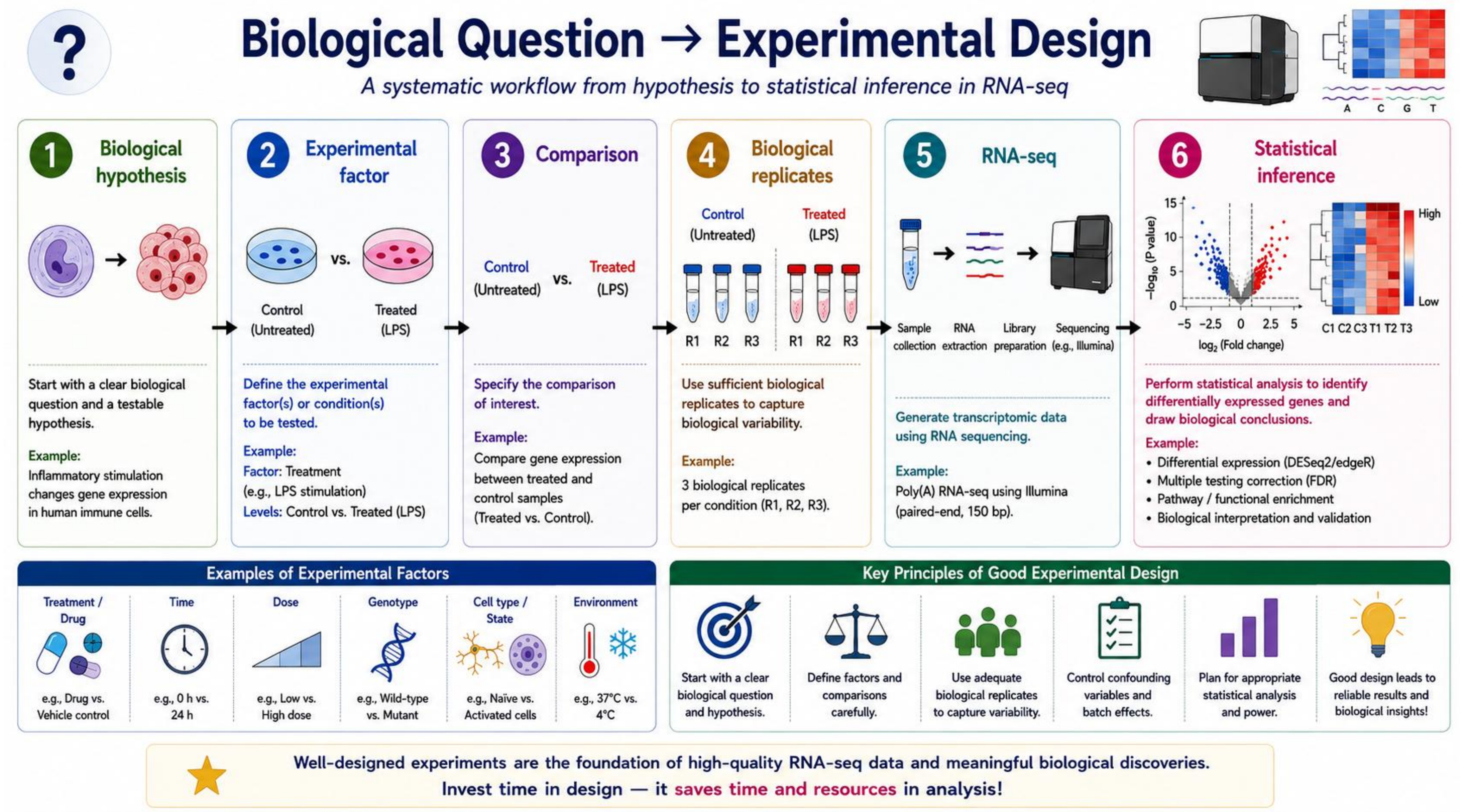
Short-Read vs Long-Read Transcriptomics

Feature	Short-Read (Illumina)	Long-Read (PacBio / ONT)
Read length	50–300 bp	1 kb – > 20 kb
Transcript coverage	Fragmented	Full-length (often)
Assembly required	Yes	Often not required
Isoform resolution	Limited / ambiguous	High / unambiguous
Complex regions	Difficult	Well resolved
Novel discovery	Limited	High
Quantification	Gene-level common	Isoform-level accurate
Best use	High throughput, cost-effective	Full-length isoforms, complexity

Long-read transcriptomics provides a more complete and accurate view of the transcriptome, enabling discovery of **full-length isoforms** and revealing the **true complexity** of gene expression.



Spatial Transcriptomics
Adding spatial coordinates to transcriptomic measurements
Spatial transcriptomics measures gene expression while preserving spatial location within the native tissue context.
1. Concept
Traditional (bulk or single-cell)
"Which genes are expressed?"
Gene expression (no spatial information)
Spatial transcriptomics
"Where are these genes expressed within the tissue?"
Gene expression
High
Low
Gene expression + Spatial position
2. How It Works
1. Tissue section on spatial array
2. Capture mRNA with spatial barcodes
3. Library prep & sequencing
4. Map reads to genome
...ATGCTAGCCTA...
...TTGACCGTAAG...
...GGCATGCTTGA...
...CTAACGGTTCA...
5. Reconstruct spatial gene expression
Each read carries a spatial barcode that identifies where in the tissue it came from.
3. Major Technologies
Spot-based (Visium, Slide-seq)
Captures RNA from spots (~55–200 µm) on a grid.
Subcellular/segmented (MERFISH, seqFISH+)
Single-molecule imaging with near-cellular resolution.
In situ capture (ST, DBiT-seq)
In situ RNA capture with barcoded probes.
High-plex imaging (COSMx, Xenium)
Multiplexed RNA imaging with high resolution.
4. Key Applications
Cell-type localization
Map where different cell types reside.
Tissue organization
Understand spatial architecture and cell–cell interactions.
Disease & cancer research
Identify spatial gene programs in tumor microenvironments.
Developmental biology
Study spatial patterns during development.
Biomarker discovery
Find spatially restricted disease biomarkers and drug targets.
5. Example: Gene Expression Across a Tissue
H&E staining
Spatial expression of Gene X
High expression
Low expression
Spatial transcriptomics reveals that Gene X is highly expressed in a specific region, not uniform across the tissue.
6. Key Advantages
Preserves native tissue context
Links gene expression to histology
Uncovers spatially restricted programs
Reveals cell–cell interactions
Improves biomarker and target discovery
Key Take-Home Message
Spatial transcriptomics bridges the gap between molecular profiling and tissue architecture—answering not only which genes are expressed, but where and in what spatial context.



Biological Question → Experimental Design
A systematic workflow from hypothesis to statistical inference in RNA-seq
A C G T
1 Biological hypothesis
Start with a clear biological question and a testable hypothesis.
Example:
Inflammatory stimulation changes gene expression in human immune cells.
2 Experimental factor
Control (Untreated) vs. Treated (LPS)
Define the experimental factor(s) or condition(s) to be tested.
Example:
Factor: Treatment (e.g., LPS stimulation)
Levels: Control vs. Treated (LPS)
3 Comparison
Control (Untreated) vs. Treated (LPS)
Specify the comparison of interest.
Example:
Compare gene expression between treated and control samples (Treated vs. Control).
4 Biological replicates
Control (Untreated)
Treated (LPS)
R1 R2 R3 R1 R2 R3
Use sufficient biological replicates to capture biological variability.
Example:
3 biological replicates per condition (R1, R2, R3).
5 RNA-seq
Sample collection
RNA extraction
Library preparation
Sequencing (e.g., Illumina)
Generate transcriptomic data using RNA sequencing.
Example:
Poly(A) RNA-seq using Illumina (paired-end, 150 bp).
6 Statistical inference
15 10 5 0
−log10 (P value)
−5 −2.5 0 2.5 5
log2 (Fold change)
High
Low
C1 C2 C3 T1 T2 T3
Perform statistical analysis to identify differentially expressed genes and draw biological conclusions.
Example:
• Differential expression (DESeq2/edgeR)
• Multiple testing correction (FDR)
• Pathway / functional enrichment
• Biological interpretation and validation
Examples of Experimental Factors
Treatment / Drug
e.g., Drug vs. Vehicle control
Time
e.g., 0 h vs. 24 h
Dose
e.g., Low vs. High dose
Genotype
e.g., Wild-type vs. Mutant
Cell type / State
e.g., Naïve vs. Activated cells
Environment
e.g., 37°C vs. 4°C
Key Principles of Good Experimental Design
Start with a clear biological question and hypothesis.
Define factors and comparisons carefully.
Use adequate biological replicates to capture variability.
Control confounding variables and batch effects.
Plan for appropriate statistical analysis and power.
Good design leads to reliable results and biological insights!
Well-designed experiments are the foundation of high-quality RNA-seq data and meaningful biological discoveries.
Invest time in design — it saves time and resources in analysis!

Technical vs Biological Variation

Separating true biological signal from unwanted sources of variation

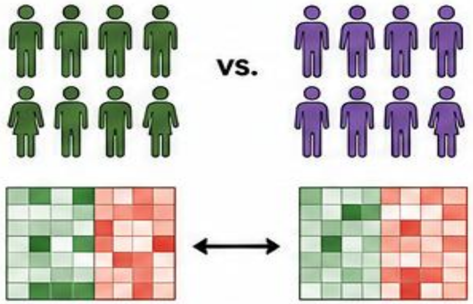
Total observed variation can be conceptualized as:

$$\text{Variation}_{\text{observed}} = \text{Variation}_{\text{biological}} + \text{Variation}_{\text{technical}} + \text{Variation}_{\text{batch}}$$

The experimental objective is to distinguish biological signal from unwanted technical variation.

1. Biological Variation

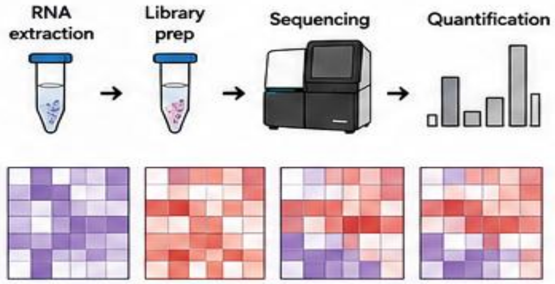
True differences between biological conditions or individuals



- Reflects real biological differences
- Due to genetics, environment, development, disease, etc.
- This is the signal of interest

2. Technical Variation

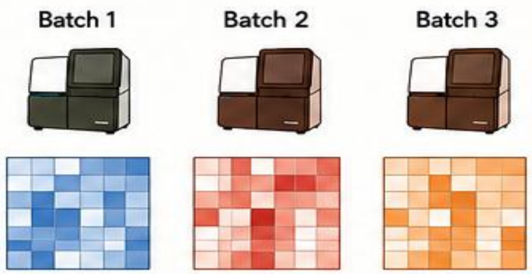
Differences introduced by experimental and measurement processes



- Due to differences in reagents, protocols, pipetting, efficiency, sequencing depth, etc.
- Can obscure the real biological signal
- Minimized by standardized protocols, randomization, and quality control

3. Batch Variation

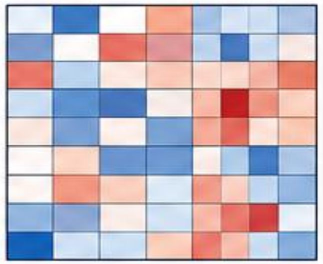
Differences arising from processing samples in different batches, times, or runs



- Caused by run date, operator, instrument, reagent lot, laboratory, etc.
- Can create systematic, non-biological differences
- Controlled by experimental design and accounted for in analysis

4. Observed Variation

What we measure in the data (a combination of all sources)



- Contains both signal (biological) and noise (technical + batch)
- Goal: extract true biological signal for meaningful interpretation

Good experimental design, replication, randomization, and appropriate statistical modeling are essential to separate biological variation from technical and batch effects.

Batch Effects

Systematic non-biological differences between batches can confound true biological signals

Potential Batch Variables

- RNA extraction date
- Library preparation batch
- Sequencing lane / flow cell
- Operator
- Reagent lot
- Tissue-processing date

These technical factors can introduce unwanted variation that masks or mimics biology.

Bad Design (Confounded)

Batch 1 = Control

All samples processed in Batch 1

Batch 2 = Treatment

All samples processed in Batch 2

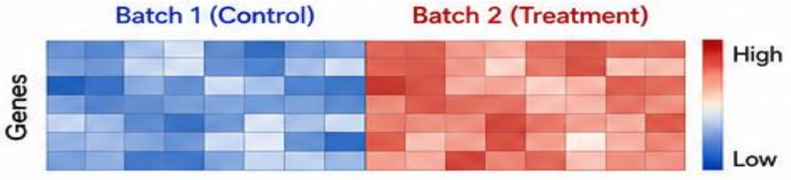
Then:
Treatment effect ≈ Batch effect

The two effects are **confounded.**

Why this is a problem?

- We cannot tell if differences are due to treatment or due to batch.
- Leads to false positives or false negatives.

Example: Expression heatmap (genes × samples)



Differences follow batches, not biology.

Good Design (Balanced)

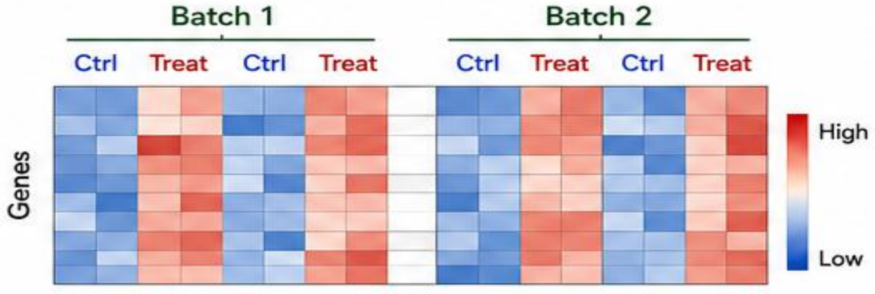
Samples from all conditions are distributed across batches.

Batch 1

Batch 2

Batch	Control	Treatment
Batch 1	✓	✓
Batch 2	✓	✓

Example: Expression heatmap (genes × samples)



Differences reflect biology, not batch.

Take-home message:

Confounding batch and treatment makes interpretation impossible.

Randomize samples, **balance** batches, and **record** batch variables to separate biology from technical noise.

Confounding Variables

Hidden factors can bias results if they are associated with the experimental condition of interest

Potential Confounding Variables

- **Sex** (e.g., male vs. female)
- **Age** (e.g., young vs. old)
- **Tissue** (e.g., brain, liver, lung)
- **Genotype** (e.g., wild-type vs. mutant)
- **Treatment** (e.g., drug vs. control)
- **Time point** (e.g., 0 h, 6 h, 24 h)
- **Batch** (e.g., extraction or library batch)
- **Sequencing lane** (e.g., lane 1 vs. lane 2)

These factors, if unevenly distributed between groups, can confound the biological signal.

Confounded Design (Problem)

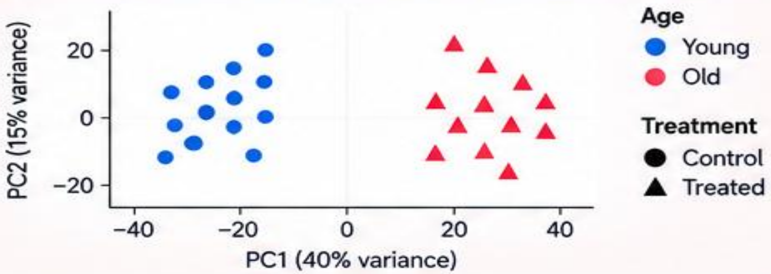
A confounder is associated with the experimental condition, making it difficult to separate their effects.

Example: All control samples are from young individuals and all treated samples are from old individuals.

Control (Young)	Treated (Old)
Age: Young Treatment: Control	Age: Old Treatment: Treated

Observed differences may be due to age, treatment, or both — and the effects are **confounded.**

Example: PCA plot (samples colored by age and shaped by treatment)



It is unclear whether the separation is due to treatment, age, or both.

Well-Designed Study (Solution)

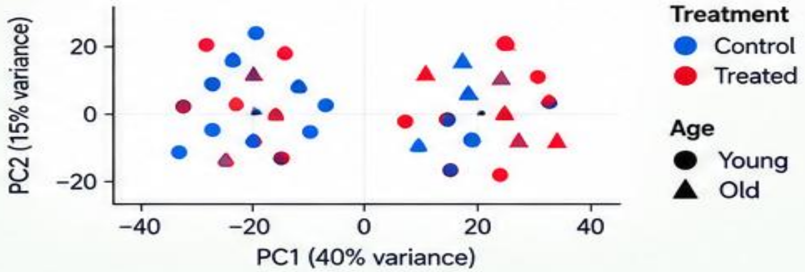
Balance or randomize confounding variables across conditions to allow proper statistical separation of effects.

Example: Include both young and old individuals in control and treated groups (balanced design).

Control (Young)	Control (Old)	Treated (Young)	Treated (Old)
Age: Young Treatment: Control	Age: Old Treatment: Control	Age: Young Treatment: Treated	Age: Old Treatment: Treated

Biological factors (e.g., treatment) can now be **separated** statistically from confounders (e.g., age).

Example: PCA plot (samples colored by treatment, shaped by age)



Separation reflects the **biological effect of interest,** not the confounding variable.

Take-home message:
A proper experimental design should ensure that important biolological factors can be **separated statistically,** by balancing, randomizing, or explicitly modeling potential confounders.

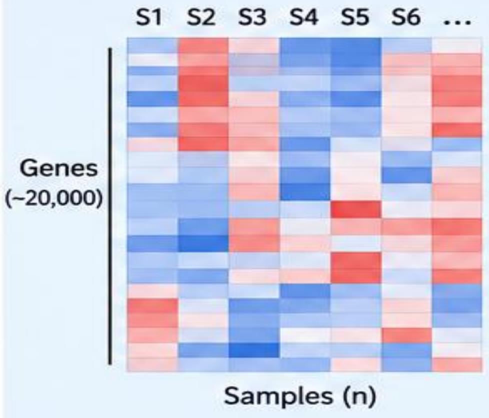
Statistical Concepts

Why RNA-seq Requires Statistical Modeling

High-dimensional count data with multiple sources of variation

RNA-seq generates

Thousands of genes × multiple samples

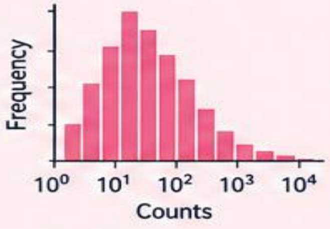


- Each entry is a count (number of reads mapped to a gene)
- Data are high-dimensional and noisy

Statistical analysis must account for:

1 Count distributions

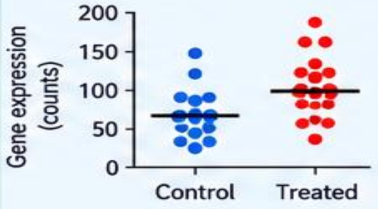
RNA-seq counts are discrete and typically overdispersed (not normal).



Modeled using distributions such as the negative binomial.

2 Biological variability

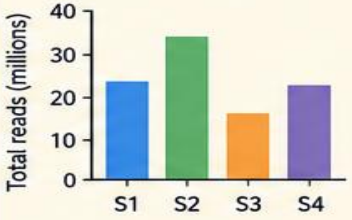
Biological replicates vary due to natural differences between individuals or conditions.



Statistical models estimate true differences while accounting for variability among replicates.

3 Library size (normalization)

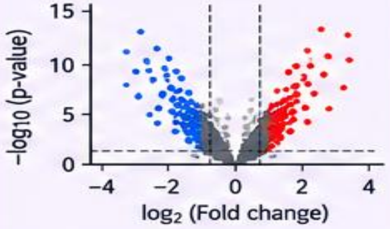
Samples have different total read counts (library sizes) that must be normalized.



Use normalization methods (e.g., size factors, TPM, TMM) to make samples comparable.

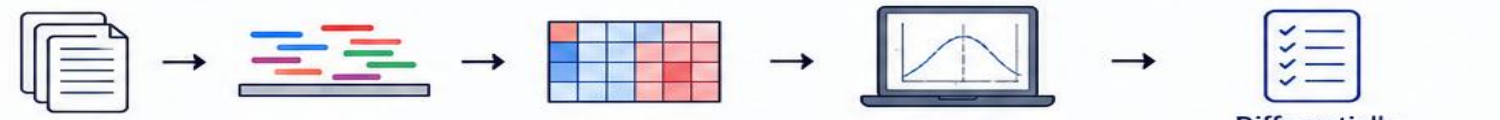
4 Multiple hypothesis testing

Thousands of genes are tested simultaneously, increasing the risk of false positives.



Adjust for multiple testing (e.g., FDR) to control false discoveries.

From Raw Data to Biological Insight



FASTQ files → Read alignment to reference genome → Count matrix (genes × samples) → Statistical modeling (e.g., DESeq2, edgeR) → Differentially expressed genes and biological interpretation

Key Take-home Message

Because RNA-seq produces thousands of gene measurements with various sources of variation, statistical modeling is essential to identify **true biological signals** and make **reliable conclusions**.

Statistical modeling turns complex RNA-seq data into meaningful biological insights.

RNA-seq Count Data

RNA-seq measures the number of reads mapped to each gene, generating integer count data

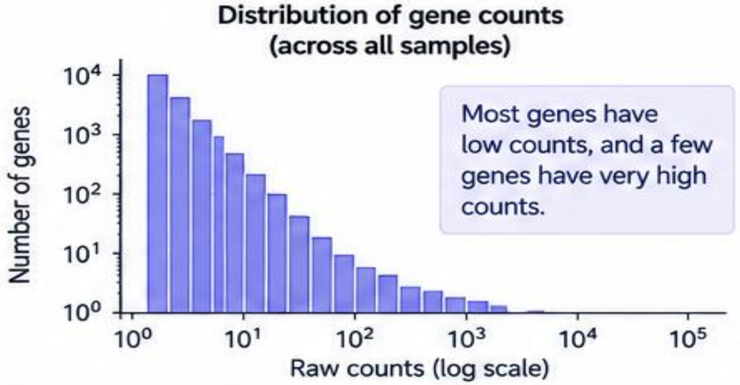
1 Count matrix (example)

RNA-seq data are typically represented as integer counts:

Gene	C1	C2	C3	T1	T2	T3
Gene A	120	135	128	520	490	540
Gene B	25	20	28	10	12	9
Gene C	300	280	310	290	310	305
Gene D	0	2	1	8	5	6
Gene E	15	18	14	60	55	70
...	...	...	...	...	...	...

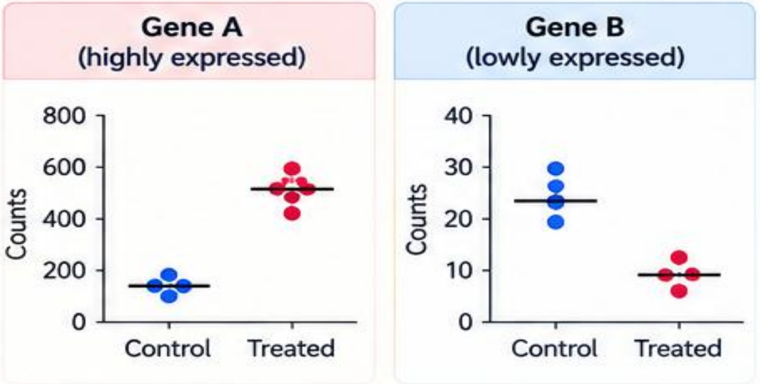
2 Distribution of raw counts

Raw RNA-seq counts are not normally **distributed.** They are typically right-skewed, with many low counts and a few very high counts.



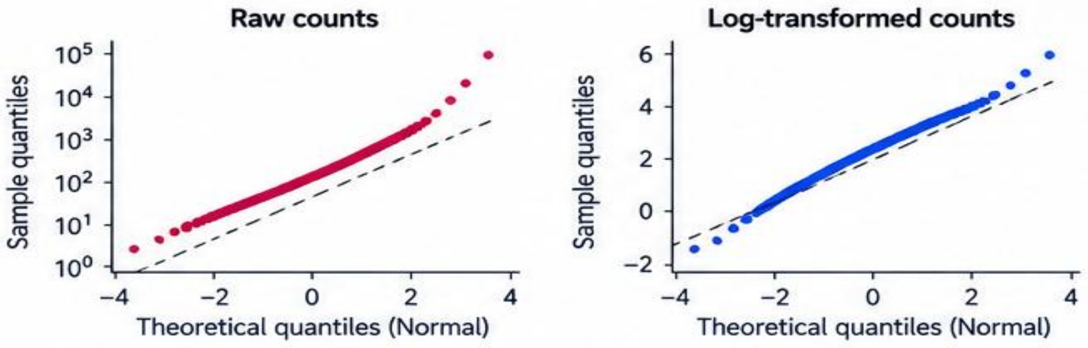
3 Example gene distributions

Different genes show different count ranges, but all exhibit non-normal (overdispersed) distributions.



4 Not normally distributed

Raw counts deviate strongly from a normal distribution.



Key points

- RNA-seq data are **integer counts** (number of reads per gene).
- Counts are **not normally distributed** (typically right-skewed).
- Many genes have low counts, a few genes have very high counts.
- Statistical methods (e.g., negative binomial models) are needed to properly model count data.
- Normalization and appropriate modeling are essential for valid inference.

Why it matters

- Standard statistical tests that assume normality (e.g., t-test) are **not** appropriate for raw counts.
- RNA-seq data require specialized methods (e.g., DESeq2, edgeR, limma-voom) that model count distributions and biological variability.
- Proper modeling helps identify truly differentially expressed genes while controlling false discoveries.
- **Transformation** (e.g., log, variance stabilizing transformation) can make the data more suitable for visualization, but statistical models are still required.

Raw RNA-seq counts are not normally distributed — statistical modeling is essential.

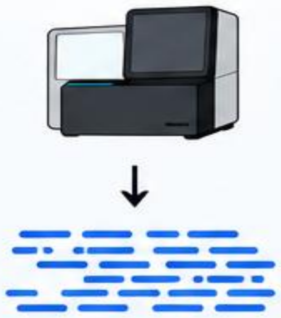
Normalization in RNA-seq

Accounting for library size and composition to make samples comparable

1. Different sequencing depths

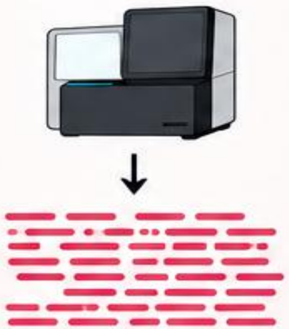
Samples can have different total numbers of sequenced reads (library sizes).

Sample A
20 million reads



20,000,000 sequenced reads

Sample B
40 million reads



40,000,000 sequenced reads

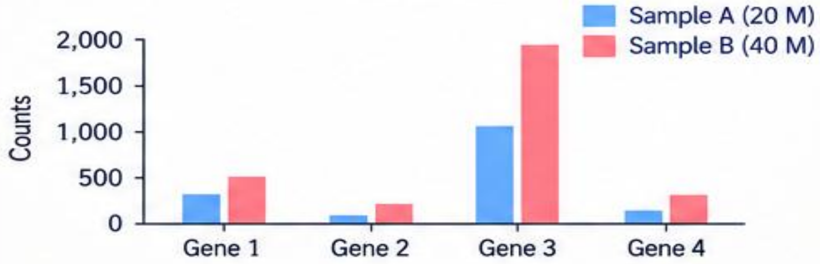
Library size (sequencing depth) can vary due to experimental design, cost, or sequencing run conditions.

2. Raw counts are not directly comparable

A gene may have a higher raw count in the sample with more total reads, even if the true expression level is the same.

Gene	Sample A (20 M reads)	Sample B (40 M reads)
Gene 1	200	400
Gene 2	50	120
Gene 3	1,000	1,900
Gene 4	25	60
...	...	...

Raw counts



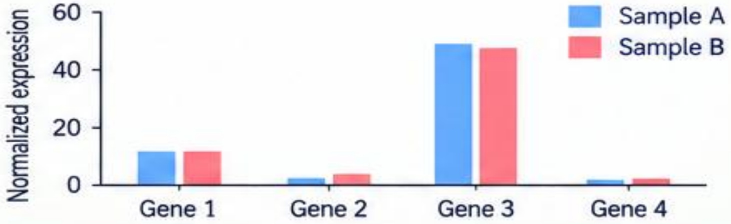
Raw counts cannot be directly compared without accounting for library size and composition.

3. After normalization

Normalization adjusts for differences in sequencing depth (and composition), making samples comparable.

Gene	Sample A (normalized)	Sample B (normalized)
Gene 1	10.0	9.8
Gene 2	2.5	3.0
Gene 3	50.0	47.5
Gene 4	1.2	1.5
...	...	...

Normalized counts
(e.g., counts per million, TPM, or DESeq2 size factors)



Normalization attempts to make samples comparable.

Common normalization methods

- Total count (CPM)
- RPKM / FPKM
- TPM
- DESeq2 size factors
- TMM (edgeR)
- Quantile normalization
- Others

Key take-home message

Normalize before comparing samples to remove the effects of sequencing depth and composition.

Goal

Ensure that observed differences reflect true biological variation, not technical differences in library size.

Differential Expression in RNA-seq

Identifying genes with significant expression changes between conditions

1. Basic comparison

Fold change compares the average expression between two conditions.

$$\text{Fold Change} = \frac{\text{Expression}_{\text{Treatment}}}{\text{Expression}_{\text{Control}}}$$

Usually reported on a $\log_2$ scale:

$$\log_2\text{FC} = \log_2\left(\frac{\text{Treatment}}{\text{Control}}\right)$$

Interpretation:

- $\log_2\text{FC} > 0$ → higher expression in treatment (upregulated)
- $\log_2\text{FC} < 0$ → lower expression in treatment (downregulated)
- $\log_2\text{FC} = 0$ → no change

2. Example calculations

Treatment	Control	Fold Change (T/C)	$\log_2$FC	Interpretation
200	100	2.0	1.0	2-fold increase (upregulated)
50	100	0.5	−1.0	2-fold decrease (downregulated)
800	100	8.0	3.0	8-fold increase
25	100	0.25	−2.0	4-fold decrease
100	100	1.0	0.0	No change

2-fold increase:

$$\log_2(2) = 1$$

Treatment expression is twice the control.

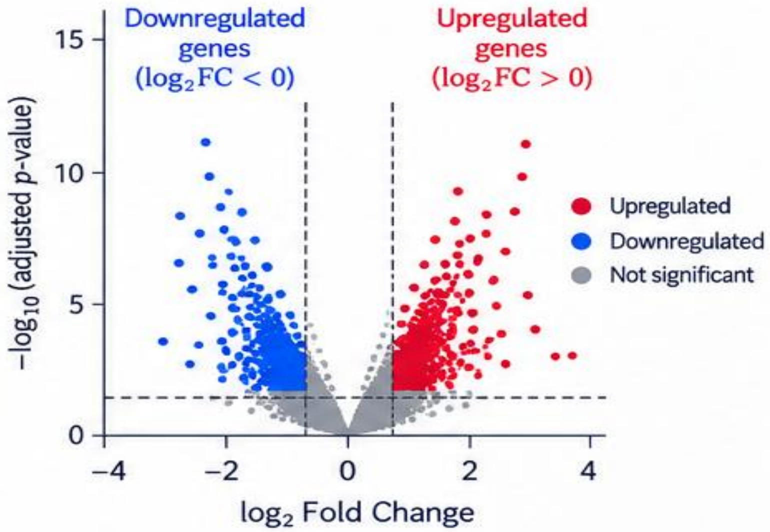
2-fold decrease:

$$\log_2(0.5) = -1$$

Treatment expression is half the control

3. Visualization: Volcano plot

Differential expression results are often visualized using a volcano plot.



4. Statistical significance

A gene is considered differentially expressed if it has:

- A large absolute $\log_2$FC (e.g., $|\log_2\text{FC}| \geq 1$)
- A statistically significant adjusted p-value (e.g., FDR < 0.05)

5. Common thresholds

Parameter	Typical threshold
$\lvert\log_2\text{FC}\rvert$	≥ 1 (2-fold change)
Adjusted p-value (FDR)	< 0.05

6. Biological interpretation

- Upregulated genes may be activated in the treatment condition.
- Downregulated genes may be repressed in the treatment condition.
- Differential expression analysis helps identify genes and pathways involved in the biological response.

Take-home message: Differential expression combines the **magnitude of change ($\log_2$FC)** and **statistical significance** (adjusted p-value) to identify genes with true expression differences between conditions.

Multiple Testing Problem in RNA-seq

When testing thousands of genes, many will appear significant by chance

1 Testing many genes

In RNA-seq, we test each gene for differential expression.

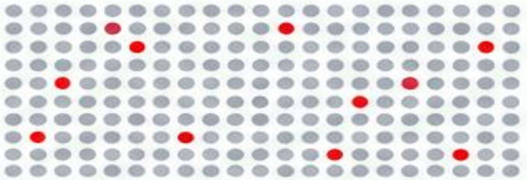
Gene	p-value
Gene 1	0.001
Gene 2	0.034
Gene 3	0.412
Gene 4	0.028
Gene 5	0.760
...	...
Gene 20,000	0.049

We perform ~20,000 statistical tests (one per gene).

2 Chance of false positives

If we use a significance threshold of $p < 0.05$, then even if there are no true differences, **5% of tests** will be significant by chance.

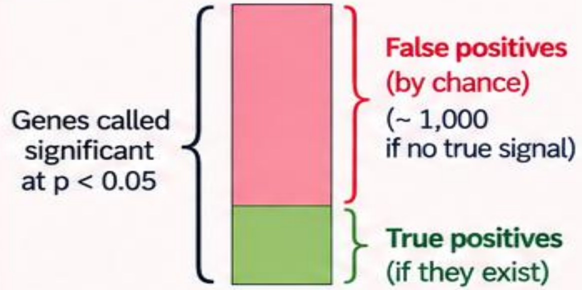
Expected false positives
$= 0.05 \times 20{,}000$
$=$ **1,000 genes**



True non-significant gene ($p \geq 0.05$)
False positive by chance ($p < 0.05$)

3 Observed vs expected

Among the significant results ($p < 0.05$), many may be false positives.



Using $p < 0.05$ alone can lead to a large number of false discoveries when testing thousands of genes.

4 False Discovery Rate (FDR)

RNA-seq therefore uses multiple testing correction, most commonly the **False Discovery Rate (FDR)** approach (Benjamini–Hochberg).

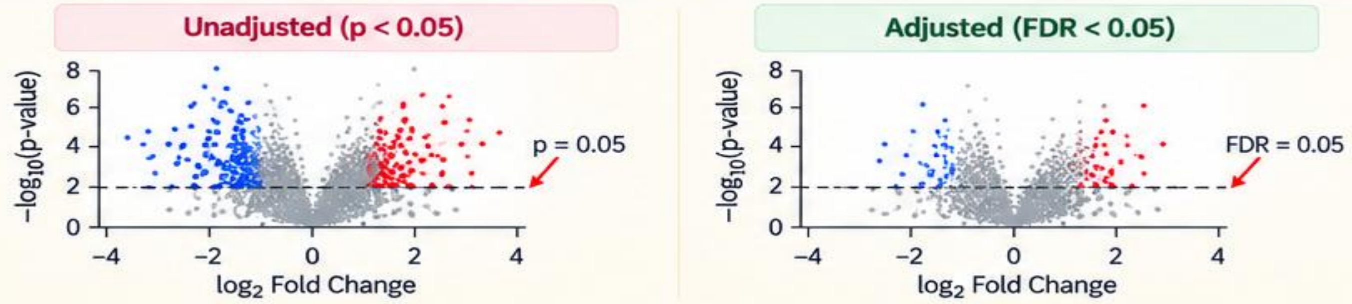
FDR is the expected proportion of false positives among the genes called significant.

Example:

FDR < 0.05

Among the genes called significant, we expect < 5% to be false positives.

Effect of multiple testing correction

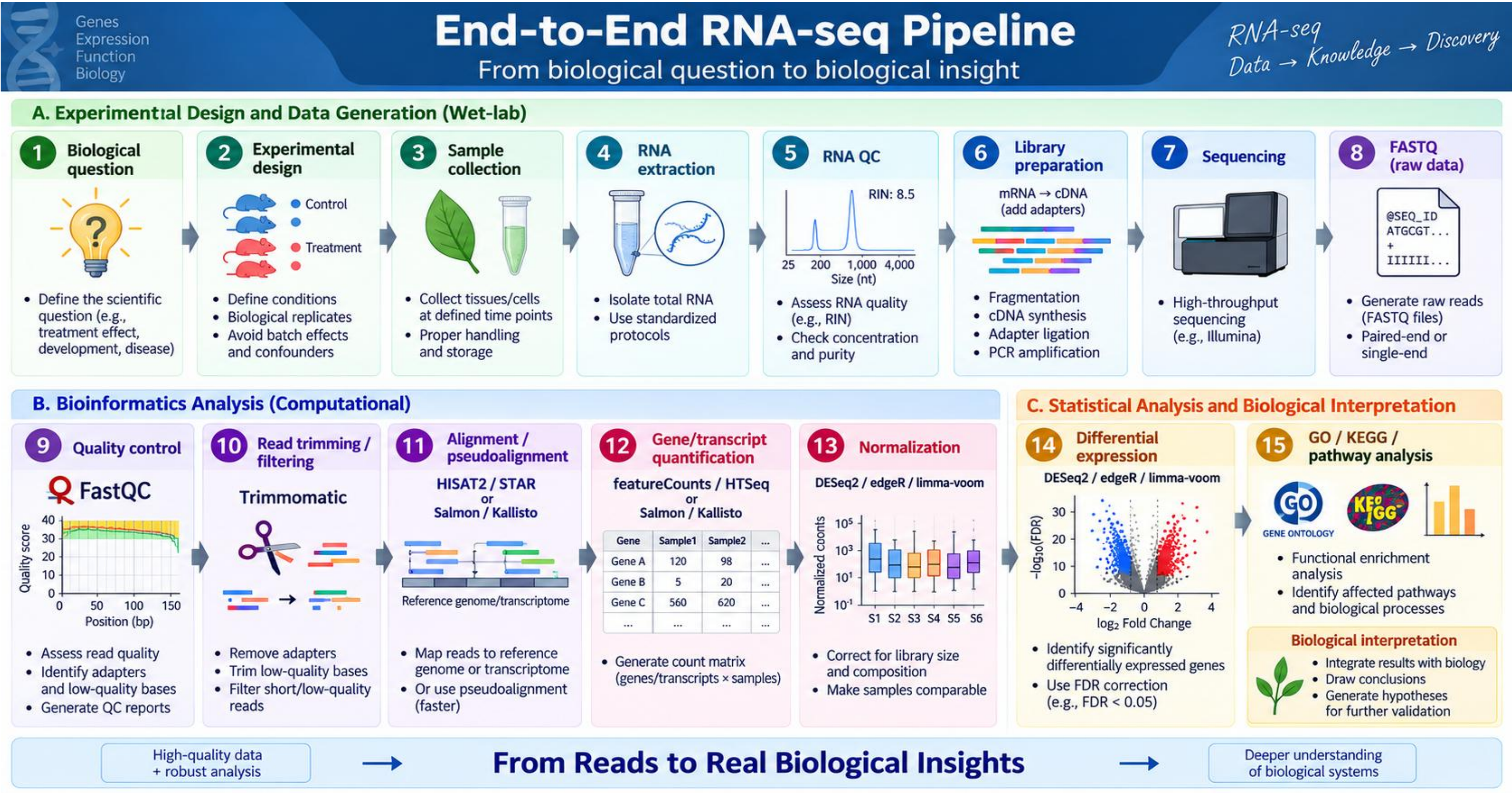


Key take-home messages

1. Testing thousands of genes increases the chance of false positives.
2. Using $p < 0.05$ alone can lead to many spurious significant genes.
3. RNA-seq analysis uses FDR correction (e.g., Benjamini–Hochberg).
4. A common threshold is FDR < 0.05.
5. FDR helps identify a reliable list of differentially expressed genes with a controlled false discovery rate.

Take-home message: **When testing 20,000 genes at $p < 0.05$, many can appear significant by chance.**
Therefore, use FDR correction (e.g., FDR < 0.05) in RNA-seq analysis.

Complete Scientific Workflow



What Makes a Good RNA-seq Experiment?

Reliable results come from good biology, good technology, and good statistics

Better design
Better data
Better biology

Biological quality

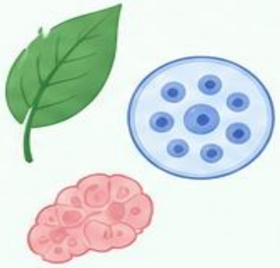
Start with a clear and relevant biological question

1 Appropriate biological question

- Well-defined and testable
- Relevant to biology, disease, development, environment, etc.

2 Representative samples

- Proper selection of tissues/cells
- Capture biological variability (e.g., different conditions, time points, individuals)
- Avoid selection bias



3 Adequate biological replication

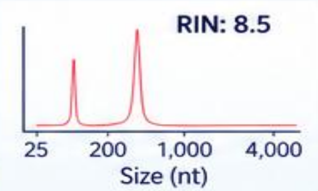
- Typically ≥ 3 biological replicates per condition
- Increases statistical power and robustness

Technical quality

Generate high-quality, reproducible data

1 High-quality RNA

- Intact RNA (e.g., RIN ≥ 7)
- Free from contamination (DNA, protein, inhibitors)
- Consistent extraction method



2 Appropriate library preparation

- Use suitable library type (e.g., poly(A), rRNA depletion, strand-specific, single-cell)
- Follow standardized protocols
- Minimize technical variation

3 Sufficient sequencing depth

- Depends on study goals
  - Gene-level analysis: ~20–30 million reads/sample
  - Isoform/single-cell: higher depth
  - Lowly expressed genes: deeper sequencing

More depth → better detection of lowly expressed genes and isoforms

Statistical quality

Design and analyze properly to obtain reliable results

1 Randomization

- Randomly assign samples to groups
- Avoid systematic biases (e.g., processing order, sequencing lane)

Randomized sample processing

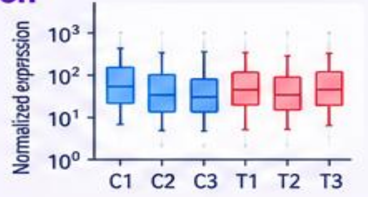
2 Balanced batches

- Distribute samples from different conditions across batches/lanes
- Avoid confounding of biological condition with batch

	Batch 1	Batch 2	Batch 3
Control	● ●	● ●	● ●
Treatment	● ●	● ●	● ●

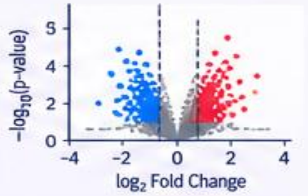
3 Appropriate normalization

- Account for library size and composition differences
- Use established methods (e.g., DESeq2, edgeR, TMM, TPM)



4 Multiple-testing correction

- Test thousands of genes simultaneously
- Control false discoveries using FDR (e.g., Benjamini–Hochberg)
- Typically use FDR < 0.05



Key takeaway: A good RNA-seq experiment integrates strong biological design, high technical quality, and rigorous statistical analysis to generate reliable and biologically meaningful results.

Good data leads to good science!

Design
Experiment
Data
Analysis
Biological Insight

Take-Home Message

Good science starts at the bench — not at the computer

Plan well
Generate good data
Get meaningful answers

Poor experimental input

Poor sampling
(non-representative samples)

No biological replicates
(no estimate of variability)

Treatment | Batch 1 | Batch 2

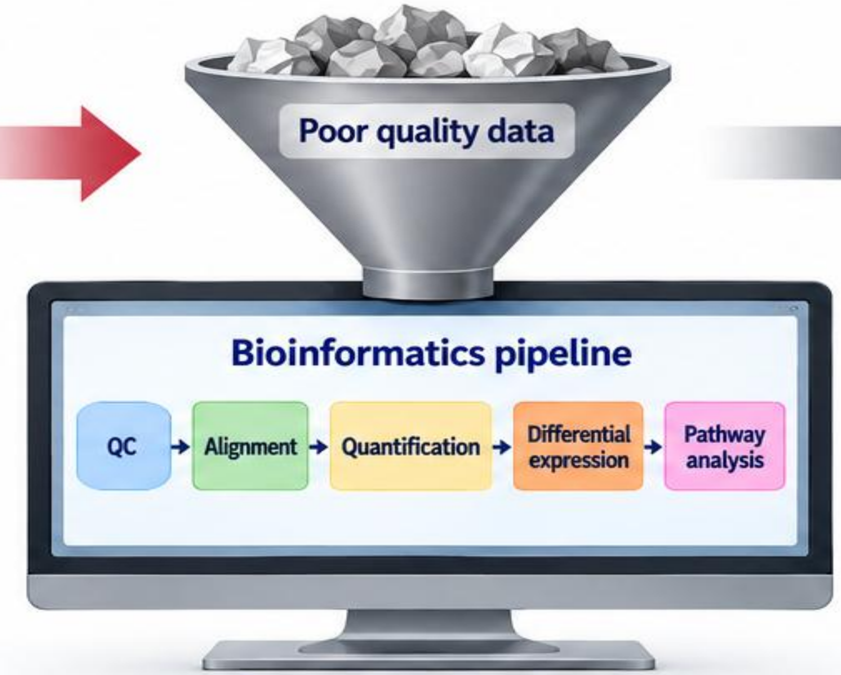
Confounded treatment and batch
(systematic bias)

Poor RNA quality
(degraded RNA, low RIN)

Incorrect experimental design
(wrong controls, insufficient randomization, etc.)

Garbage in → garbage out

A sophisticated bioinformatics pipeline cannot rescue poorly designed or low-quality experiments.



Even the best pipeline...

Unreliable biological conclusions

Spurious differential expression results

Misleading functional and pathway analysis

False biological interpretation

Wasted time, resources and money

Loss of confidence in results

The **quality of the biological experiment** determines the **quality of the computational conclusion.**

Good data leads to real discoveries!

Knowledge today
Better biology tomorrow

Key References for RNA-seq and Single-cell Studies

Foundational papers shaping modern transcriptomics and single-cell genomics

Read
Learn
Apply
Advance Science

1 Wang *et al.* (2009)

RNA-Seq: a revolutionary tool for transcriptomics

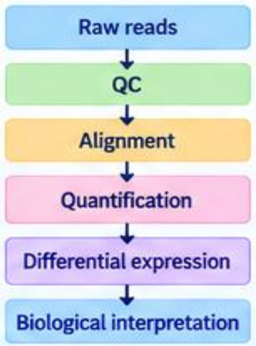
Nature Reviews Genetics
10: 57–63 (2009)

- Seminal review introducing RNA-seq and its potential to revolutionize transcriptomics.
- Covers key applications, advantages over microarrays, and future perspectives.

2 Conesa et al. (2016)

A survey of best practices for RNA-seq data analysis

Genome Biology
17: 13 (2016)

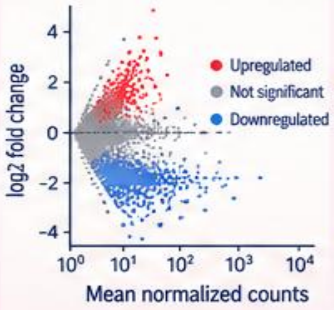


- Comprehensive guidelines for RNA-seq data analysis.
- Covers experimental design, data processing, statistical analysis, and reproducibility.
- Widely cited best practices paper.

3 Love et al. (2014)

Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2

Genome Biology
15: 550 (2014)

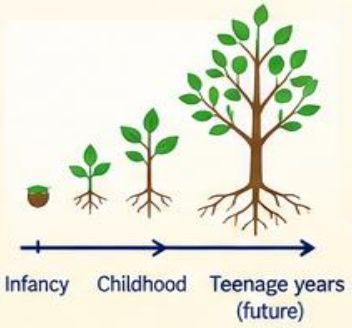


- Introduces DESeq2 for differential expression analysis.
- Uses shrinkage estimation for improved fold change and dispersion estimates.
- Robust and widely used in the field.

4 Stark *et al.* (2019)

RNA sequencing: the teenage years

Nature Reviews Genetics
20: 631–656 (2019)

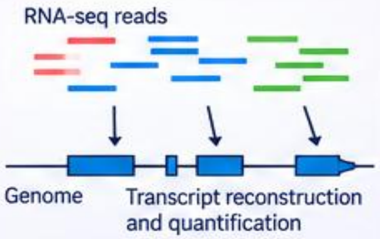


- Reflects on a decade of RNA-seq technology and applications.
- Discusses current challenges and future directions.
- Excellent perspective on where RNA-seq stands and where it is headed.

5 Mortazavi *et al.* (2008)

Mapping and quantifying mammalian transcriptomes by RNA-Seq

Nature Methods
5: 621–628 (2008)

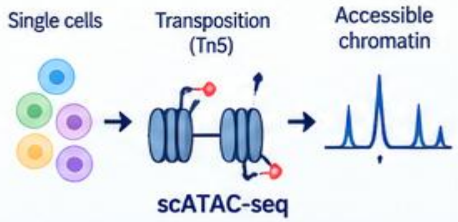


- One of the first studies to demonstrate genome-wide transcriptome profiling using RNA-seq.
- Introduces methods for mapping and quantifying transcripts.
- Showed the power of RNA-seq to discover novel transcripts and isoforms.

6 Buenrostro *et al.* (2015)

Single-cell chromatin accessibility

Nature Methods
12: 715–718 (2015)

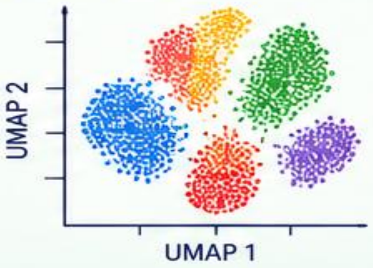


- Introduces single-cell ATAC-seq for profiling chromatin accessibility at single-cell resolution.
- Enables linking regulatory elements to cell types and states.
- Highly influential in single-cell genomics.

7 Stuart & Satija (2019)

Integrative single-cell analysis

Nature Reviews Genetics
20: 257–272 (2019)



- Comprehensive review of computational approaches for single-cell data integration and analysis.
- Covers dimensionality reduction, clustering, trajectory inference, and data integration.
- A key resource for modern single-cell transcriptomics analysis.

These landmark studies provide the foundation for RNA-seq and single-cell analysis.
Read, explore, and build on them to design better experiments and generate deeper biological insights.

Good papers drive great science!

Basic Linux Commands and WSL Concept

Essential skills for Bioinformatics and Computational Biology

Learn Linux
Work Smarter
Do Better Science

1 Basic Linux Commands

Common commands you will use every day

Task	Command	Example	Description
Check system info	uname -a	uname -a	Show system information
Current directory	pwd	pwd	Print working directory
List files	ls	ls -lh	List files and directories (-l: long format, -h: human readable)
Change directory	cd	cd /home/user	Change directory
Go to home	cd ~	cd ~	Go to home directory
Go to parent directory	cd ..	cd ..	Move up one directory
Create directory	mkdir	mkdir data	Create a new directory
Remove file	rm	rm file.txt	Delete a file
Remove directory	rm -r	rm -r folder	Delete a directory (recursively)
Copy file/directory	cp	cp file.txt /path/	Copy files or directories
Move/rename	mv	mv old.txt new.txt	Move or rename files/directories
View file content	cat	cat file.txt	Display file content
View file (scroll)	less	less file.txt	Browse file content (press q to quit)
View beginning	head	head -n 10 file.txt	Show first 10 lines
View end	tail	tail -n 10 file.txt	Show last 10 lines
Search text	grep	grep "gene" file.txt	Search for a pattern in files
Find files	find	find . -name "*.fastq"	Find files by name
Check disk space	df	df -h	Show disk space usage
Check directory size	du	du -sh folder	Show directory size
File permissions	chmod	chmod +x script.sh	Change file permissions
View processes	ps	ps aux | less	View running processes
Kill a process	kill	kill 1234	Terminate a process (by PID)
Download file	wget	wget URL	Download files from internet
Network test	ping	ping google.com	Check network connectivity
Clear terminal	clear	clear	Clear the terminal screen

Useful shortcuts

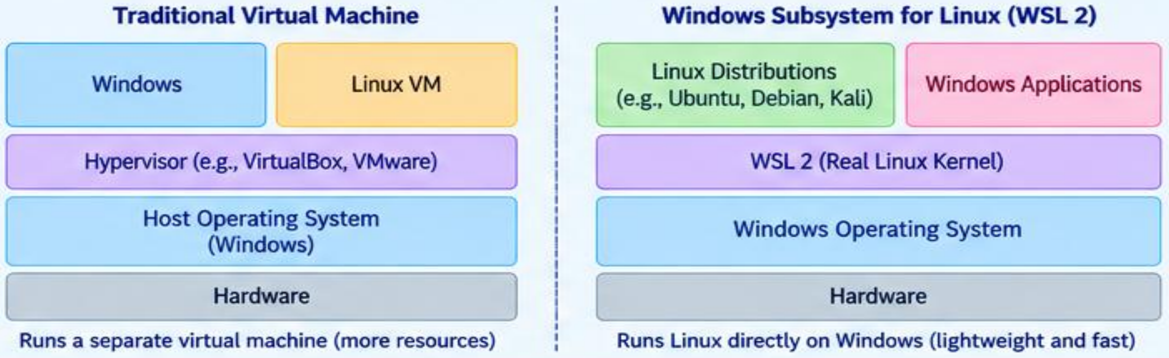
Ctrl + C	Stop a running command
Ctrl + Z	Suspend a command
Ctrl + D	Exit terminal (logout)
Tab	Auto-complete file/command
↑ / ↓	Recall previous commands
!cmd	Run last command starting with cmd
history	Show command history

2 What is WSL?

Windows Subsystem for Linux (WSL) lets you run a Linux environment on Windows.

- WSL allows you to run a full Linux distribution (e.g., Ubuntu) directly on Windows without a virtual machine.
- You can use Linux command-line tools, bioinformatics software, and scripts alongside your Windows files.
- WSL 2 provides a real Linux kernel, better performance, and full system call compatibility.

How WSL Works



3 Install and Use WSL (Windows 10/11)

Step-by-step guide

Step	Action	Command
1	Open PowerShell as Administrator	powershell
2	Install WSL	wsl --install
3	Restart your computer	
4	Launch Linux (e.g., Ubuntu)	wsl
5	Set default version to WSL 2 (optional)	wsl --set-default-version 2
6	List installed distributions	wsl --list --verbose
7	Access Windows files from WSL	cd /mnt/c
8	Access Linux files from Windows	\\wsl$\Ubuntu\home\<user>

Popular Linux Distributions for WSL

- **Ubuntu** (Most popular, recommended)
- **Debian** (Stable and lightweight)
- **Kali Linux** (Security and penetration testing)
- **Others** (AlmaLinux, CentOS, etc.)

Tips

- Use a terminal like Windows Terminal (recommended).
- You can install bioinformatics tools (e.g., conda, BWA, HISAT2) in WSL.
- Your Windows files are available under /mnt/ (e.g., /mnt/c/Uers/).
- WSL 2 is faster and more compatible than WSL 1.
- Ideal for bioinformatics, programming, and reproducible research.

Same Linux Power. More Possibilities.

Windows + Linux
A Better Research Workflow